

72nd BPSC PRELIMS

प्रहार

Test Series

Test- 31/15

FULL LENGTH TEST

Solution

English



BPSC TEST-31/15 (SOLUTION)

Q 1. Ans. D

Explanation:

- **Statement 1 is correct:** The 16th Finance Commission (FC) recommended retaining the states' vertical share in the divisible pool of central taxes at 41% for the award period of 2026–2031, maintaining the status quo established by the 15th FC.
- **Statement 2 is correct:** In a major shift to the horizontal devolution formula (how money is distributed among states), the 16th FC introduced a new parameter called “**Contribution to GDP**” with a **10% weightage**. This replaced the previous “Tax and Fiscal Efforts” criterion (which held a 2.5% weight) to explicitly reward states for their economic output and efficiency.
- **Statement 3 is correct:** Breaking from decades of tradition, the 16th FC **completely discontinued** the Article 275 Revenue Deficit Grants, as well as sector-specific and state-specific grants. The Commission shifted toward a more performance-oriented fiscal model, limiting its grant allocations primarily to Local Bodies and Disaster Management.
- **Additional Facts:**
 - ✓ **Timeline and Leadership:** Chaired by Dr. Arvind Panagariya (former Vice Chairman of NITI Aayog), the 16th FC submitted its final report to President Droupadi Murmu on November 17, 2025. The recommendations were officially tabled in Parliament on February 1, 2026, covering the five-year period from FY 2026-27 to 2030-31.
 - ✓ **Revised Horizontal Devolution Weights:** The final formula for distributing taxes among states is:
 - **Income Distance (Per Capita GSDP Distance):** 42.5% (reduced from 45%)
 - ◇ Income distance holds the maximum weightage at 42.5%, followed by **Population at 17.5%**, and **10% each for Demographic Performance, Area, Forest & Ecology, and Contribution to GDP.**
 - **Population (2011):** 17.5% (increased from 15%)
 - **Area:** 10% (reduced from 15%)
 - **Demographic Performance:** 10% (reduced from 12.5%)
 - **Forest:** 10% (retained, but now includes both overall area and recent increases in forest cover)
 - **Contribution to GDP:** 10% (Newly introduced)
- ✓ **Refined Demographic Parameter:** While the 15th FC measured “Demographic Performance” using the Total Fertility Rate (TFR), the 16th FC redefined it to measure population growth strictly between 1971 and 2011, rewarding states that successfully controlled population growth during that historical window.
- ✓ **Fiscal Discipline:** The Commission firmly recommended capping state fiscal deficits at 3% of GSDP and explicitly mandated that states completely discontinue the practice of off-budget borrowings, bringing all liabilities transparently onto their main budgets.
- ✓ **Disaster Management:** The 16th FC recommended adding **heatwaves and lightning** to the official list of notified national disasters, making affected states eligible for compensation from the State Disaster Response Fund (SDRF).

Q 2. Ans. C

Explanation:

- **Statement 1 is correct:** In the Union Budget 2025-26, the credit guarantee cover under the Credit Guarantee Scheme for Startups (CGSS) was officially doubled from ₹10 crore to ₹20 crore per eligible borrower. This enhancement, accompanied by a reduced guarantee fee of 1% for 27 priority sectors, is designed to help capital-intensive startups secure easier access to collateral-free formal credit.
- **Statement 2 is correct:** The Budget significantly expanded the agricultural credit ecosystem by increasing the loan limit under the Modified Interest Subvention Scheme (MISS) from ₹3 lakh to ₹5 lakh for short-term loans accessed through the Kisan Credit Card (KCC). This ensures that farmers, including those in the dairy and fisheries sectors, can adequately meet rising cultivation and post-harvest costs.
- **Additional Facts:**
 - ✓ **NIFTEM:** The Union Budget 2025-26 formally announced the establishment of

a **National Institute of Food Technology, Entrepreneurship and Management (NIFTEM)** in Hajipur, Bihar.

- ✓ **Agriculture Credit Target:** The government fixed a record-high agricultural credit target of **₹32.50 lakh crore** for FY 2025-26, dedicating a sub-target of ₹5 lakh crore entirely to animal husbandry, dairying, and fisheries.
- ✓ **Startup Extensions:** A new **₹10,000 crore Fund of Funds** was established specifically to expand venture capital support for early-stage startups. Furthermore, the sunset clause for startups eligible to claim a tax holiday under Section 80-IAC was extended by five years, allowing startups incorporated before **March 31, 2030**, to avail of the benefits.
- ✓ **MSME Support:** Beyond startups, the credit guarantee cover for micro and small enterprises was also increased from ₹5 crore to ₹10 crore. A new customised Credit Card scheme was launched to provide ₹5 lakh in credit specifically to micro enterprises registered on the Udyam portal.

Q 3. Ans. A

Explanation:

- **Justice P.P. Navlekar Committee (a - 2):** Formed by the Union Home Ministry to conduct a comprehensive, evidence-based study of unnatural demographic changes arising from illegal immigration. The committee focuses primarily on border districts and rapidly expanding metropolitan areas.
- **Justice Ranjana Prakash Desai Commission (b - 1):** Former Supreme Court judge Ranjana Prakash Desai is currently serving as the Chairperson of the recently constituted **8th Central Pay Commission**. The commission is tasked with recommending revisions to the salary structure, allowances, and pensions of Central Government employees and pensioners.
- **Prof. Alakh Sharma Committee (c - 4):** In June 2026, the Ministry of Statistics set up a 22-member expert panel headed by Prof. Alakh Sharma to examine new metrics for India's rapidly evolving labour market, specifically focusing on capturing data regarding gig/platform workers, moonlighting, and the informal sector.
- **Justice S. Ravindra Bhat Task Force (d - 3):** Appointed by the Supreme Court of India, this National Task Force was formed to investigate the

predominant causes of student suicides in Higher Education Institutions (HEIs) and recommend frameworks for adequate mental health support.

● Additional Facts:

- ✓ **Bihar & Labour Migration (Alakh Sharma Context):** Prof. Alakh Sharma is the Director of the Institute for Human Development (IHD). A major focus of the IHD's research is analysing longitudinal data on the massive seasonal and circular labour migration originating from **Bihar** to more industrialised states, a critical driver of the state's rural economy.
- ✓ **SC Task Force Findings on HEIs:** The interim report submitted by the Justice S. Ravindra Bhat panel revealed that an alarming **65%** of surveyed higher education institutions in India currently lack access to any full-time mental health professionals or counsellors.
- ✓ **8th CPC Members:** Alongside Chairperson Ranjana Prakash Desai, the 8th Central Pay Commission includes members **Pankaj Jain** and **Prof. Pulak Ghosh**.

Q 4. Ans. C

Explanation:

- According to the official data released by the Department for Promotion of Industry and Internal Trade (DPIIT) in 2025, **Singapore** remained the top investing country for the financial year 2024-25.
- Singapore led with a massive **30% share** of the total FDI equity inflows (amounting to approx USD 15 billion).
- **Mauritius** was the second-largest source with a 17% share, followed by the **United States** in third place with an 11% share.
- **Additional Facts:**
 - ✓ **Top Recipient State:** **Maharashtra** retained its position as the top recipient state, capturing the highest share (39%) of total FDI equity inflows, followed by Karnataka (13%) and Delhi (12%).
 - ✓ **Top Sector:** The **Services Sector** (including banking, financial, and insurance) attracted the highest FDI equity inflows at 19%, overtaking the Computer Software & Hardware sector (16%).
 - ✓ **Total FDI Growth:** India's gross FDI inflows reached a three-year high of **USD 81.04 billion** in FY 2024-25, marking a 14% year-on-year increase from the previous fiscal year.

Q 5. Ans. B

Explanation:

- **Statement 1 is incorrect:** According to the NFHS-6 factsheets, India's Total Fertility Rate (TFR) has completely stabilised but has not dropped to 1.8. It remained steady at 2.0, which is exactly the same as the TFR recorded during NFHS-5, effectively maintaining the replacement-level demographic transition.
 - **Statement 2 is correct:** The survey highlights a substantial and positive decline in chronic childhood undernutrition. At the national level, stunting (low height-for-age) among children under five years fell from 35.5% in NFHS-5 to 29.3% in NFHS-6.
 - **Statement 3 is correct:** Under the government's push for "data harmonisation," the NFHS-6 factsheet controversially dropped 30 indicators that were previously tracked in NFHS-5. The total number of indicators was reduced from 131 to 101. Key omitted metrics include cancer screening (cervical, breast, oral), the sex ratio at birth, and sanitation data.
 - **Additional Facts:**
 - ✓ Based on the NFHS-6 data, **Bihar** registered an excellent performance in reducing the share of underweight children. The state saw a decline of 5.3%, with the proportion of underweight children dropping from 41.0% (NFHS-5) to 35.7% (NFHS-6).
 - ✓ The Ministry of Health and Family Welfare designated the **International Institute for Population Sciences (IIPS)**, Mumbai, as the sole nodal agency to coordinate and conduct NFHS-6.
 - ✓ **Institutional Deliveries:** Highlighting a massive structural victory for public healthcare, the national rate of institutional deliveries surged past the ninety-percent mark, rising from 88.6% in NFHS-5 to 90.6% in NFHS-6.
 - ✓ **Rising Obesity:** While undernutrition metrics improved, the survey exposed a rapidly growing non-communicable disease risk. The share of women classified as overweight or obese rose sharply to 30.7% (up from 24.0% in NFHS-5 and 12.6% in 2005-06).
 - ✓ **Health Financing:** The share of households covered by some form of health insurance or state-sponsored financing scheme recorded a steep increase from 41% to 60% nationwide.
- Poverty Index (MPI) measures acute poverty across three equally weighted dimensions: **Health, Education, and Standard of Living** (each carrying a 1/3 weightage).
- **Statement 2 is incorrect:** A person is classified as multidimensionally poor if they are deprived in 33.33% (**one-third**) or more of the weighted indicators. A deprivation score of 20% to 33.3% classifies a person as "vulnerable to poverty," while a score of 50% or more indicates "severe multidimensional poverty."
 - **Statement 3 is incorrect:** 'Ownership of a bank account' is **not** included in the 10 standard indicators of the Global MPI published by the UNDP/OPHI. The global standard of living indicators are: cooking fuel, sanitation, drinking water, electricity, housing, and assets.
 - ✓ Bank accounts and maternal health are additional indicators used exclusively in India's National MPI prepared by NITI Aayog.
 - **Additional Facts:**
 - ✓ **Publishers:** The Global MPI is jointly published annually by the United Nations Development Programme (UNDP) and the Oxford Poverty and Human Development Initiative (OPHI).
 - ✓ **India's National MPI (NITI Aayog):** While the Global MPI uses exactly 10 indicators, India's National MPI uses **12 indicators**. NITI Aayog localised the index by adding two specific metrics: **Maternal Health** (under the Health dimension) and **Bank Accounts** (under the Standard of Living dimension).
 - ✓ **Bihar & National MPI Data:** According to the latest finalised NITI Aayog National MPI Progress Report, **Bihar** recorded the highest proportion of multidimensionally poor population in India at **33.76%**. However, Bihar also registered the sharpest absolute drop in poverty, with the headcount ratio declining by 18.13 percentage points from its baseline of 51.89%.
 - ✓ **Bihar's Top/Bottom Districts:** Within Bihar, **Kishanganj, Araria, and Madhepura** recorded the highest percentage of multidimensionally poor populations, while **Patna** recorded the lowest.

Q 6. Ans. A

Explanation:

- **Statement 1 is correct:** The Global Multidimensional

Q 7. Ans. C

Explanation:

- The Union Cabinet approved Phase IV of the Pradhan Mantri Gram Sadak Yojana (PMGSY-IV) in

September 2024. The core target of this phase is the construction of **62,500 km** of all-weather rural roads.

- This new infrastructure is designed to provide connectivity to **25,000** previously unconnected rural habitations.

- The phase is being implemented over a five-year period from FY 2024-25 to 2028-29, backed by a total financial outlay of **₹70,125 crore**.

- **Additional Facts:**

- ✓ **Eligibility Criteria (Census 2011):** The scheme targets habitations with a population of **500+ in plains, 250+ in North-Eastern/Himalayan states and special category areas** (Tribal Schedule V, Aspirational Districts/Blocks), and **100+ in Left Wing Extremism (LWE) affected districts**.
- ✓ Across all phases of PMGSY, **Bihar ranks first** in the country for the highest number of habitations connected, having successfully linked **31,287 habitations** to the rural road network.
- ✓ **The financial sharing ratio is 60:40** (Centre:State) for general states, **90:10** for North-Eastern and Himalayan States, and **100% centrally funded** for Union Territories without a legislature.
- ✓ To ensure maximum coverage of marginalised groups, PMGSY-IV habitations are prioritised in convergence with the Dharti Aaba Janjatiya Gram Utkarsh Abhiyan (for tribal/PVTG populations) and PM-AJAY (for SC populations).
- ✓ **Digital Monitoring:** The progress and quality of roads are tracked in real-time via **OMMAS** (Online Management, Monitoring, and Accounting System), **e-MARG** (for post-construction maintenance), and the newly launched **"PMGSY Gram Sadak Survey App"** for identifying optimal road alignments

Q 8. Ans. B

Explanation:

- **Statement 1 is correct:** Petroleum Refinery Products hold the maximum weightage in the Index of Eight Core Industries at **28.04%**.
- **Statement 2 is correct:** Fertilisers hold the lowest weightage in the index at just **2.63%**.
- **Statement 3 is incorrect:** The Eight Core Industries comprise **40.27%** of the total weight of items included in the Index of Industrial Production (IIP).

- **Additional Facts:**

- ✓ **Decreasing Order of Weightage:** The sequence from highest to lowest weight is: **Refinery Products (28.04%) > Electricity (19.85%) > Steel (17.92%) > Coal (10.33%) > Crude Oil (8.98%) > Natural Gas (6.88%) > Cement (5.37%) > Fertilisers (2.63%)**.
- ✓ **Base Year:** The current base year for calculating the Index of Eight Core Industries is **2011-12**.
- ✓ **Publishing Agency:** The ICI is compiled and released monthly by the **Office of the Economic Adviser (OEA)**, Department for Promotion of Industry and Internal Trade (DPIIT), under the Ministry of Commerce & Industry.
- ✓ **Difference from IIP:** While the ICI is published by the DPIIT, the broader Index of Industrial Production (IIP) is published by the **National Statistical Office (NSO)** under the Ministry of Statistics and Programme Implementation (MoSPI).
- ✓ The Ministry of Statistics and Programme Implementation (MoSPI) revised the **base year of the IIP to 2022-23** (replacing the old 2011-12 base)
- ✓ Under this new 2026 series, MoSPI officially discontinued the use of the Wholesale Price Index (WPI) as a deflator for value-based outputs, replacing it with the newly introduced **Output Producer Price Index (Output PPI)**.

Q 9. Ans. B

Explanation:

- The **National Health Claims Exchange (NHCX)** is a digital gateway developed jointly by the National Health Authority (NHA) and the Insurance Regulatory and Development Authority of India (IRDAI).
- Its primary purpose is to create a centralised, interoperable platform that enables the seamless exchange of claims-related information between healthcare providers (hospitals), insurance companies, and Third-Party Administrators (TPAs).
- By standardising the data flow, the NHCX aims to completely digitise the health insurance claims process, drastically reducing settlement times, administrative costs, and the wait times for patients during hospital discharge.
- **Additional Facts:**
 - ✓ **Parent Mission:** The NHCX operates under the broader umbrella of the **Ayushman Bharat**

Digital Mission (ABDM), which is tasked with creating a seamless online platform for India's digital health ecosystem.

- ✓ **Key Identifier (ABHA):** For the NHCX to function efficiently, it integrates with the **Ayushman Bharat Health Account (ABHA)**, a unique **14-digit identification number** used to digitally link and store a patient's medical records across different healthcare providers.
- ✓ **Bima Sugam:** While NHCX focuses on claims, IRDAI has also been rolling out **Bima Sugam**, an e-commerce-like centralised marketplace designed to act as a one-stop shop for buying, selling, and servicing all types of insurance policies in India.
- ✓ **Nodal Agencies:** The **National Health Authority (NHA)** is the apex body responsible for implementing both the Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) and the ABDM. IRDAI, headquartered in Hyderabad, is the statutory body regulating the insurance and reinsurance industries.

Q 10. Ans. D

Explanation:

- **Statement A is correct:** According to the 18th edition of the India State of Forest Report (ISFR) 2023, the total forest and tree cover of India is **8,27,357 sq km**, which constitutes exactly **25.17%** of the country's total geographical area. (Forest cover alone is 21.76%, and tree cover is 3.41%).
- **Statement B is correct:** In terms of absolute area, the top three states with the largest forest cover are **Madhya Pradesh** (77,073 sq km), followed by **Arunachal Pradesh** and **Chhattisgarh**.
- **Statement C is correct:** When measuring forest cover as a percentage of a region's total geographical area, Lakshadweep is the highest overall (91.33%). However, strictly among States, **Mizoram** has the highest percentage at **85.34%**.
- **Additional Facts:**
 - ✓ **Publication:** The ISFR is a biennial report published by the Forest Survey of India (FSI). The 2023 report (released in late 2024) utilised remote sensing data sourced from ISRO's **LISS-III sensor** on the IRS Resourcesat satellite.
 - ✓ **Maximum Increase vs Decrease:** **Chhattisgarh** recorded the highest absolute increase in forest and tree cover (+684 sq km). Conversely, **Madhya Pradesh** recorded the maximum

loss in forest and tree cover (-612.41 sq km), despite retaining the top overall spot.

- ✓ **Mangrove Cover:** India's total mangrove cover stands at 4,991.68 sq km (0.15% of the total geographical area). The 2023 report noted a **net decrease of 7.43 sq km** in mangrove cover since the 2021 assessment.
- ✓ **Agroforestry Inclusion:** For the first time in the history of the ISFR, trees grown under agroforestry (orchards, bamboo, and palm plantations outside traditional forest lands) were explicitly delineated and included in the assessment.
- ✓ **Bihar Specific Fact (ISFR 2023):** Bihar emerged as the state with the **second-highest increase in forest cover outside the Recorded Forest Area (RFA)**, adding **106.8 sq km**. It was second only to Gujarat (+241 sq km) in this specific metric.

Q 11. Ans. B

Explanation:

- **Pair 1 is correctly matched:** In the World Press Freedom Index 2026 published by Reporters Without Borders (RSF), India dropped six places to rank **157th** out of 180 countries.
- **Pair 2 is correctly matched:** According to the World Happiness Report 2026, India improved its position slightly, ranking **116th** out of 147 countries.
- **Pair 3 is incorrectly matched:** In the Climate Change Performance Index (CCPI) 2026 released by Germanwatch, India dropped out of the top 10 for the first time in several years, falling 13 places to rank **23rd** globally.
- **Pair 4 is incorrectly matched:** In the Henley Passport Index, India's passport rank was 80th globally, enabling visa-free or visa-on-arrival access to 56 countries.
- **Additional Facts:**
 - ✓ **Global Innovation Index (GII) 2025:** Published by the World Intellectual Property Organisation (WIPO), India climbed to the **38th** position out of 139 economies, maintaining its top rank among lower-middle-income economies.
 - ✓ **Corruption Perceptions Index (CPI) 2025:** Released by Transparency International, India secured the **91st** rank out of 182 countries, marking an improvement of five places from its previous ranking of 96.
 - ✓ **Global Gender Gap Index 2025:** Published by the World Economic Forum (WEF), India

slipped two spots to rank **131st** out of 148 countries. Within South Asia, Bangladesh remained the top performer.

- ✓ **Top Performers:** Norway maintained its 1st rank in the Press Freedom Index, Finland topped the World Happiness Report for the ninth consecutive year, Denmark remained the least corrupt nation in the CPI, and Singapore held the first spot on the Henley Passport Index.

Q 12. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Justice Sheel Nagu served as the Chief Justice of the **Punjab and Haryana High Court** immediately prior to his elevation to the Supreme Court.
- **Pair 2 is incorrectly matched:** Justice Sanjeev Sachdeva was serving as the Chief Justice of the **Madhya Pradesh High Court**, not Jammu & Kashmir and Ladakh.
- **Pair 3 is incorrectly matched:** Justice Arun Palli was the Chief Justice of the **High Court of Jammu & Kashmir and Ladakh**, not Madhya Pradesh.
- **Pair 4 is correctly matched:** Justice Shree Chandrashekhar served as the Chief Justice of the **Bombay High Court** before being elevated to the apex court.
- **Additional Facts:**
 - ✓ On June 2, 2026, a batch of five new judges took the oath of office for the Supreme Court, comprising Justices Sheel Nagu, Shree Chandrashekhar, Sanjeev Sachdeva, Arun Palli, and V. Mohana.
 - ✓ **Direct Elevation:** Senior Advocate V. Mohana was **directly elevated** from the Bar to the Supreme Court bench. She is only the **11th advocate** in Indian judicial history to be directly elevated in this manner, and the **second woman** to achieve this distinction, after Justice Indu Malhotra.
 - ✓ **Women in the Judiciary:** With Justice V. Mohana's appointment, the total number of women judges appointed in the history of the Supreme Court of India reached **12**.

Q 13. Ans. C

Explanation:

- **Statement 1 is correct:** The Unified Pension Scheme (UPS) was introduced as an **optional** pension scheme operating under the broader National Pension System

(NPS) framework. It is not mandatory; eligible central government employees have the option to choose between the NPS and the UPS.

- **Statement 2 is incorrect:** The UPS is regulated and administered by the **Pension Fund Regulatory and Development Authority (PFRDA)**, not the Insurance Regulatory and Development Authority of India (IRDAI).
- **Statement 3 is incorrect:** Under the UPS, the employee's contribution remains at **10%** of Basic Pay + Dearness Allowance (DA). However, the Central Government's total contribution is **18.5%** (comprising a matching 10% plus an additional 8.5% allocated to a common pool corpus to fund the guaranteed benefits). It is not 20% each.
- **Statement 4 is correct:** New Central Government recruits joining on or after April 1, 2025, must explicitly exercise their option to join the UPS within **30 days** of joining the service. If no choice is made within this window, the NPS automatically becomes their default option.
- **Additional Facts:**
 - ✓ **Assured Pension Amount:** Retirees who complete a minimum of **25 years** of qualifying service are guaranteed a monthly pension equal to **50% of their average basic pay** drawn over the last 12 months prior to superannuation.
 - ✓ **Minimum Guaranteed Pension:** Employees who have completed at least **10 years** of service are guaranteed a minimum pension payout of **₹10,000 per month**.
 - ✓ **Family Pension:** In the event of the pensioner's death, their eligible spouse or family will receive an assured family pension equivalent to **60%** of the pension that was being drawn by the retiree.
 - ✓ **Inflation Indexation:** Unlike the NPS, both the assured pension and the family pension under the UPS are fully indexed to inflation, receiving Dearness Relief (DR) calculated using the **All India Consumer Price Index for Industrial Workers (AICPI-IW)**.
 - ✓ **Effective Date:** The Unified Pension Scheme officially came into effect on **April 1, 2025**.
 - ✓ **First State Adopter:** **Maharashtra** became the first state in India to officially approve the implementation of the UPS for its state government employees.

Q 14. Ans. B

Explanation:

- The **PARIVARTAN** (Programme for Accelerated

Renewal and Incentivisation of Vehicle Assets for Reducing Transport Air Pollution and Network Emissions) scheme is designed specifically to combat vehicular air pollution in the National Capital Region (NCR).

- Approved by the Union Cabinet in June 2026 with a total financial outlay of ₹9,585 crore, its primary objective is to aggressively phase out older, highly polluting commercial trucks and buses operating in Delhi-NCR and replace them with cleaner **Bharat Stage (BS)-VI-compliant** or fully **electric vehicles**.

● **Additional Facts:**

- ✓ **Implementing Agency:** While the overarching guidelines were approved by the Ministry of Housing and Urban Affairs (MoHUA), the ground-level execution of the scheme is being handled by the **Ministry of Road Transport and Highways (MoRTH)**.
- ✓ **Funding Body:** The capital for the scheme is being routed through the **National Capital Region Planning Board (NCRPB)**, reflecting a unified, inter-state approach to tackling the region's air quality crisis.
- ✓ **Incentive Package:** To make the transition financially viable for commercial drivers, the scheme offers a 10-year Motor Vehicle Tax concession, complete registration fee waivers, a 5% interest subvention on new vehicle loans, and a guaranteed minimum 8% manufacturer (OEM) discount.
- ✓ **Participating States:** The states of **Haryana, Rajasthan, Uttar Pradesh, and the NCT of Delhi** have all officially notified the required tax concessions to operationalise the scheme within their respective borders.
- ✓ **Digital Integration:** Delivery of the scheme is entirely digitised, directly interfacing with existing national automotive portals like **VAHAN** (registration), **V-Scrap** (scrappage), and **DigiELV** to ensure transparent and seamless fund transfers.

Q 15. Ans. C

Explanation:

- The **Swachhata Hi Seva (SHS) 2025** campaign was a nationwide cleanliness drive jointly organised by the **Ministry of Housing and Urban Affairs (MoHUA)** (overseeing urban areas) and the Department of Drinking Water and Sanitation under the **Ministry of Jal Shakti** (overseeing rural areas).
- The campaign is an annual event that runs from

September 17 to October 2, culminating on **Gandhi Jayanti (Swachh Bharat Diwas)**.

- The official theme for the 2025 edition was **"Swachhotsav"**, a concept designed to blend the spirit of festive celebrations with the civic responsibility of sanitation.

● **Additional Facts:**

- ✓ **Cleanliness Target Units (CTUs):** A core operational pillar of the 2025 campaign was the identification, mapping, and transformation of CTUs, difficult, neglected garbage spots and legacy waste dumps into clean, usable public spaces.
- ✓ **SafaiMitra Suraksha Shivirs:** The campaign placed a high priority on the welfare of frontline sanitation workers by organising nationwide single-window health and safety camps (Shivirs) focused on occupational healthcare and linking workers to social security schemes.
- ✓ **Mass Shramdaan:** A massive nationwide volunteering event was executed on September 25, 2025, under the banner **'Ek Din, Ek Ghanta, Ek Saath'**, which saw millions of citizens, officials, and volunteers participate simultaneously in physical cleaning drives.
- ✓ **Digital Monitoring:** To ensure transparency and track real-time progress, the entire campaign, from the creation of cleanliness events to the final closure of remediated garbage spots, was monitored through a dedicated centralised SHS IT Portal.

Q 16. Ans. B

Explanation:

- **Size of Economy (Nominal GDP):** India's nominal GDP officially overtook Japan in the 2025-26 period, making it the **4th largest economy** in the world (behind the US, China, and Germany). By Purchasing Power Parity (PPP), India remains 3rd.
- **Milk Production:** India continues to hold the **1st position** globally, consistently accounting for nearly a quarter of total global milk production.
- **Mobile Phone Manufacturing:** Driven heavily by the Production-Linked Incentive (PLI) scheme, India has emerged as the world's **2nd largest** mobile phone manufacturer by volume. In 2025, smartphones surpassed traditional goods to become the single largest exported product category from India.
- **Renewable Energy Installed Capacity:** According to the Renewable Energy Statistics 2026 released by the International Renewable Energy Agency (IRENA),

India surpassed Brazil to become the **3rd largest** country globally in terms of total renewable energy installed capacity (behind China and the USA).

● **Additional Facts:**

- ✓ **Automobile Market:** India is the **3rd largest** automobile market in the world, having surpassed Japan.
- ✓ **Startup Ecosystem:** India retains its position as the **3rd largest** startup ecosystem globally, trailing only the US and China, with 126 recognised unicorns as of early 2026.
- ✓ **Steel Production:** India is the world's **2nd largest** producer of crude steel.
- ✓ **Aviation Market:** India is currently the **3rd largest** domestic aviation market globally and the fastest-growing major market.

Q 17. Ans. D

Explanation:

- The **CapaCITIES** (Capacity Building for Low Carbon and Climate Resilient City Development in India) programme is officially funded by the **Embassy of Switzerland to India and Bhutan** (through the Swiss Agency for Development and Cooperation).
- The initiative aims to strengthen the capacities of Indian cities to mainstream low-carbon, climate-resilient development into their urban governance and master planning processes.
- The project is implemented by a consortium led by ICLEI South Asia, with the **National Institute of Urban Affairs (NIUA)** acting as the central knowledge partner.
- **Additional Facts:**
 - ✓ **Core Objective:** The programme directly supports India's commitment to achieving **net-zero emissions by 2070** by aligning local city-level climate action with broader national initiatives like Mission LiFE, the Smart Cities Mission, and AMRUT.
 - ✓ **Participating Cities:** The project has provided dedicated technical and financial support to urban local bodies in cities such as Ahmedabad, Rajkot, Vadodara, Udaipur, Siliguri, Coimbatore, Tiruchirappalli, and Tirunelveli.
 - ✓ **Recent Milestone:** In July 2026, the programme marked its 10-year legacy with a massive national event titled 'Scaling Urban Climate Resilience' held at the India Habitat Centre in New Delhi, bringing together officials from over 30 cities and 6 state governments.

- ✓ During the July 2026 convention, the NIUA and its partners officially launched a **Net-Zero Climate Resilient Cities Methodology Toolkit**, an energy transition guidebook for urban local bodies, and a comprehensive white paper on climate finance.

Q 18. Ans. B

Explanation:

- Following their historic and maiden ODI World Cup victory in 2025, the **Indian Women's Cricket Team** was officially nominated for the prestigious **Laureus World Team of the Year 2026** award.
- They also became the **first-ever women's cricket team** to be shortlisted for this globally recognised sporting honour.
- The Harmanpreet Kaur-led side was nominated alongside international sporting heavyweights, including the England Women's Football Team, Paris Saint-Germain (PSG), the McLaren Formula 1 team, and the European Ryder Cup Team.
- **Additional Facts:**
 - ✓ **Ultimate Winner:** The **Paris Saint-Germain (PSG)** football club ultimately clinched the 2026 Laureus World Team of the Year award.
 - ✓ The Indian Women's Team's nomination was only the third time Indian cricket featured prominently at the Laureus Awards. Previously, **Sachin Tendulkar** won the Laureus Sporting Moment Award (2000–2020) for his 2011 World Cup victory lap, and **Rishabh Pant** was nominated for the Comeback of the Year award.
 - ✓ Outside cricket, notable Indian athletes who have received Laureus nominations in the past include javelin thrower **Neeraj Chopra** (Breakthrough of the Year) and wrestler **Vinesh Phogat** (Comeback of the Year).

Q 19. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Indian journalists Suparna Sharma and R.K. Anand (along with Natalie Obiko Pearson) won the Pulitzer Prize 2026 in the 'Illustrated Reporting and Commentary' category for their Bloomberg project "trAPPed", which highlighted the growing menace of digital surveillance and cyber fraud in India.
- **Pair 2 is correctly matched:** Soma Mandal (Mondal), a science teacher from Delhi, won the Cambridge Dedicated Teacher Awards 2026 for the

South Asia region. She also went on to become the first educator from India to win the global title for her climate-focused curriculum.

- **Pair 3 is correctly matched:** Major Abhilasha Barak, the first woman combat helicopter pilot of the Indian Army, was awarded the UN Military Gender Advocate of the Year Award (for 2025, presented in June 2026) for her exemplary service with the UN Interim Force in Lebanon (UNIFIL).
- **Pair 4 is correctly matched:** The Lokmat Sur Jyotsna National Music Award 2026 was conferred upon renowned Hindustani classical vocalists Sumitra Guha and Laxman Krishnarao (Lakshman Pandit) for their exceptional, lifelong contributions to Indian classical music.
- **Additional Facts:**
 - ✓ **UN Peacekeeping (India's Legacy):** Major Abhilasha Barak is the third Indian peacekeeper to receive the UN Military Gender Advocate of the Year Award. The previous Indian recipients were Suman Gawani (2019, South Sudan) and Radhika Sen (2023, Democratic Republic of the Congo).
 - ✓ The Pulitzer Prizes, administered by Columbia University, are regarded as among the highest global honours in journalism, literature, and music composition.
 - ✓ **Global Teacher Recognition:** The Cambridge Dedicated Teacher Award is presented by Cambridge University Press & Assessment to honour teachers worldwide who make exceptional contributions to shaping students' futures.

Q 20. Ans. B

Explanation:

- **Statement 1 is correct:** Following its expansion, the BRICS+ grouping represents approximately 46% of the global population.
- **Statement 2 is incorrect:** The expanded BRICS does not account for over 50% of global GDP. It represents roughly 29% to 30% of global nominal GDP and approximately 37% of global GDP based on Purchasing Power Parity (PPP).
- **Statement 3 is correct:** With the strategic inclusion of major Middle Eastern energy powerhouses, the expanded BRICS bloc now accounts for roughly 43.1% of global crude oil production and over 44% of proven global oil reserves.
- **Additional Facts:**
 - ✓ On January 1, 2024, the original five-member group

formally expanded to include **Egypt, Ethiopia, Iran, and the United Arab Emirates (UAE)**.

- ✓ At the 2024 Kazan summit and throughout 2025/2026, BRICS introduced a “partner country” tier to accommodate growing interest without immediately granting full membership. Nations like **Vietnam** have actively joined under this partner framework.
- ✓ **Core Financial Institutions:**
 - **New Development Bank (NDB):** Headquartered in Shanghai, China, the NDB was established in 2015 with an initial subscribed capital of \$50 billion to fund infrastructure and sustainable development projects.
 - **Contingent Reserve Arrangement (CRA):** Created in 2014 with an initial funding pool of \$100 billion, this mechanism is designed to provide rapid liquidity support to member states experiencing short-term balance-of-payments pressures.
- ✓ **Critical Metals Dominance:** Beyond hydrocarbons, the expanded bloc dominates the green transition supply chain, holding roughly **79% of global aluminium production** and **77% of palladium production**.

Group of Twenty (G20):

- **Population:** Accounts for roughly 66% (two-thirds) of the world's total population.
- **GDP & Trade:** Represents approximately 85% of the global nominal GDP and over 75% of global international trade.

Shanghai Cooperation Organisation (SCO):

- **Population:** With its expanded 10-member roster (including Iran and Belarus), the SCO is the world's largest regional organisation by demographic scope, covering approximately 40% to 42% of the global population.
- **GDP:** It accounts for around 23% of the world's nominal GDP and approximately 36% of global GDP measured by PPP.

Group of Seven (G7):

- **Population:** Accounts for only about 10% of the global population.
- **GDP:** Despite its smaller demographic footprint, this bloc of advanced economies represents nearly 45% of global nominal GDP and roughly 28% of global GDP based on PPP.

European Union (EU):

- **Population:** The 27-member bloc accounts for less than 6% of the global population.

- **GDP:** It generates roughly **one-sixth** (~16-17%) of the global nominal GDP, making it the third-largest economic bloc by PPP (approx. 13.7%), trailing behind China and the U.S.

Association of Southeast Asian Nations (ASEAN):

- **Population:** The 10-member bloc accounts for roughly 8.5% of the global population.
- **GDP:** It represents approximately 3.5% to 4% of global nominal GDP, though it remains one of the fastest-growing regional economic integration zones in the world.

Q 21. Ans. A

Explanation:

- In January 2026, the State Board of Wildlife (SBWL) of **Chandigarh** officially declared the '**Striped Tiger**' (*Danaus genutia*) as the Union Territory's State Butterfly.
- The decision was taken during a meeting chaired by the Punjab Governor and UT Administrator, Gulab Chand Kataria.
- **Additional Facts:**
 - ✓ **Characteristics:** The Striped Tiger butterfly features striking tawny/orange wings marked with broad black veins and margins dotted with white spots, resembling tiger stripes.
 - ✓ **Habitat:** It is abundantly found across Chandigarh, particularly in the Sukhna Wildlife Sanctuary and the Sector 26 Butterfly Park.
 - ✓ **Protection Status:** In India, it is listed under Schedule I of the Wildlife Protection Act, 1972, granting it the highest level of legal protection.

Q 22. Ans. C

Explanation:

- According to the recently released **Unified District Information System for Education Plus (UDISE+)** data, India achieved a significant milestone by officially recording over **1 crore school teachers** for the first time.
- **Additional Facts:**
 - ✓ **Improved Pupil-Teacher Ratio (PTR):** Due to the surge in teacher numbers, the national PTR remained well within the **National Education Policy (NEP) 2020-recommended** benchmark of 30:1. The ratios now stand at approximately 10:1 at the Foundational stage, 12:1 at the Preparatory stage, 17:1 at the Middle stage, and 21:1 at the Secondary stage.
 - ✓ **Female Representation:** The proportion of female teachers in the school workforce has

seen a steady upward trajectory, climbing to over **54.2%** of the total teaching staff, reflecting growing gender parity in the profession.

- ✓ **Decline in Dropouts:** The report noted a sharp fall in dropout rates across key educational stages. The dropout rate declined to roughly 2.3% at the preparatory stage and 8.2% at the secondary stage. However, ensuring students complete secondary education remains the system's biggest challenge.
- ✓ **School Count & Rationalisation:** India currently operates a massive network of over **14.7 lakh schools** serving nearly 25 crore students. Encouragingly, the number of 'single-teacher schools' has declined by about 6%, and 'zero-enrolment schools' decreased by 38% due to better resource allocation.
- ✓ **Infrastructure Boost:** The data shows substantial progress in basic amenities, with over **99% of schools** now having drinking water facilities, nearly 97% featuring dedicated girls' toilets, and over 93% connected to the electricity grid.
- ✓ **What is UDISE+?:** It is one of the world's largest Education Management Information Systems (EMIS) maintained by the **Ministry of Education**, serving as the primary data repository for evidence-based policymaking in the school sector.

Q 23. Ans. D

Explanation:

- **Blue Revolution (a - 1):** The Blue Revolution is associated with the significant increase in the production of fish and marine products, aiming to maximize the economic potential of aquaculture and fisheries.
- **Golden Revolution (b - 2):** The Golden Revolution refers to a period of rapid growth and development in the agricultural sector specifically concerning horticulture crops (fruits and vegetables) and honey production.
- **White Revolution (c - 3):** The White Revolution is associated with a massive increase in milk and dairy production. It was designed to make the nation self-sufficient in milk and to improve the economic conditions of dairy farmers through cooperative structures.
- **Yellow Revolution (d - 4):** The Yellow Revolution is associated with a strategic surge in the production of edible oilseeds (such as mustard, sunflower,

groundnut, and soybean) to achieve domestic self-reliance in cooking oils.

● **Additional Facts:**

- ✓ **Green Revolution:** Food grains (specifically wheat and rice).
- ✓ **Silver Revolution:** Eggs and poultry production.
- ✓ **Red Revolution:** Meat and tomato production.
- ✓ **Round Revolution:** Potato production.
- ✓ **Pink Revolution:** Onion production, pharmaceuticals, and prawn cultivation.
- ✓ **Grey Revolution:** Fertilizer production.
- ✓ **Brown Revolution:** Leather, cocoa, and non-conventional energy production.
- ✓ **Black Revolution:** Petroleum production.
- ✓ **Silver Fiber Revolution:** Cotton production.
- ✓ **Golden Fiber Revolution:** Jute production.
- ✓ **Evergreen Revolution:** Overall sustainable agricultural development.

Q 24. Ans. C

Explanation:

- The **Pradhan Mantri Skilling and Employability Transformation through Upgraded ITIs (PM-SETU)** scheme is a major centrally sponsored initiative launched by the Ministry of Skill Development and Entrepreneurship (MSDE). Its primary objective is to modernise and transform **1,000 Government Industrial Training Institutes (ITIs)** across India into state-of-the-art, industry-aligned skilling hubs.
- **Additional Facts:**
 - ✓ **Hub-and-Spoke Model:** The scheme employs a “hub-and-spoke” framework to optimise resources and expertise. This structure includes:
 - **200 Hub ITIs:** These serve as central “Centres of Excellence,” providing advanced infrastructure, innovation labs, and specialised training facilities.
 - **800 Spoke ITIs:** These are linked to the hub institutes, benefiting from shared resources, curriculum support, and faculty development.
 - ✓ **Industry-Led Governance:** A defining characteristic of PM-SETU is the shift toward an “industry-managed” model. Each cluster is overseen by a **Special Purpose Vehicle (SPV)**, where industry partners play a pivotal role in curriculum design, practical training, and ensuring that the skills imparted align with current market demands.

- ✓ **Objective:** By upgrading laboratories, introducing digital classrooms, and incorporating emerging technology-based courses (such as AI, robotics, and advanced manufacturing), the scheme aims to enhance the employability of Indian youth and bridge the gap between vocational training and industrial requirements.
- ✓ **Funding:** The scheme is supported by a substantial financial outlay of ₹60,000 crore, co-financed by the Central Government, State Governments, and industry partners, with additional support from the World Bank and the Asian Development Bank (ADB).

Q 25. Ans. D

Explanation:

- The Economic Survey 2025-26 reports India’s total foodgrain production for 2024-25 at 357.73 million tonnes, a record high (3,577 lakh metric tonnes), which rounds to **357.7 MMT**.
- This was an 8% increase over the previous year’s 332.30 million tonnes, the largest decade-long growth recorded in Indian agriculture.
- Rice hit a record ~150.18 MMT, and within the same Survey, horticulture output of 367.72 million tonnes actually surpassed foodgrain production for the year.

Q 26. Ans. B

Explanation:

- India’s first fully indigenous hydrogen fuel cell passenger vessel was built by **Cochin Shipyard Limited (CSL)**.
- The vessel commenced its commercial operations in **Varanasi** (on the Ganga River, National Waterway 1) in **December 2025**.
- It is owned by the Inland Waterways Authority of India (IWAI).
- **Additional Facts:**
 - ✓ **Vessel Specs:** Designed as a 24-meter catamaran for urban transit, it has a seating capacity of 50 passengers in a fully air-conditioned cabin and can operate for up to eight hours on a single hydrogen fill.
 - ✓ **Technology:** The vessel operates on a **Low-Temperature Proton Exchange Membrane (LT-PEM)** fuel cell system.
 - ✓ **Zero-Emission:** The system generates electricity by combining stored hydrogen with oxygen, producing **water** as the only byproduct, ensuring zero smoke and noise pollution.

- ✓ **Significance:** The deployment is a major milestone for the Harit Nauka initiative and the National Hydrogen Energy Mission, aligning with India's broader commitment to achieving net-zero emissions by 2070.

Q 27. Ans. D

Explanation:

- The official theme for National Science Day 2026, as announced by the Department of Science and Technology (DST), was "Women in Science: Catalysing Viksit Bharat".
- The theme was strategically chosen to emphasize the crucial role of women scientists, researchers, and innovators in driving India toward its goal of becoming a developed nation (Viksit Bharat) by 2047.
- It aligns with broader international and national efforts to bridge the gender gap in STEM (Science, Technology, Engineering, and Mathematics) fields and promote women's leadership in high-tech sectors like AI, space, and biotechnology.
- **Additional Facts:**
 - ✓ **Core Significance:** The day is celebrated annually across India on **February 28** to commemorate the discovery of the **Raman Effect**.
 - ✓ **The Discovery:** On February 28, 1928, Indian physicist **Sir Chandrasekhara Venkata Raman (C.V. Raman)** and his collaborator K.S. Krishnan discovered that when light passes through a transparent material, some of the deflected light changes wavelength due to interactions with molecular vibrations.
 - ✓ **Nobel Prize:** For this groundbreaking work proving the quantum nature of light, C.V. Raman was awarded the **Nobel Prize in Physics in 1930**, making him the first Asian to win a Nobel Prize in any scientific field.
 - ✓ **Establishment:** The Government of India, acting on a proposal by the National Council for Science and Technology Communication (NCSTC), officially designated February 28 as National Science Day in 1986. The inaugural nationwide celebration took place in 1987.

Q 28. Ans. C

Explanation:

- The **Samrat Samprati Museum** (Samrat Samprati Sangrahalay) is located at Koba Tirth in Gandhinagar, Gujarat.

- Named after Samrat Samprati, the grandson of Emperor Ashoka, who is highly revered in Jain tradition for his commitment to non-violence and the propagation of Jainism, the museum showcases the rich historical, cultural, and spiritual legacy of the Jain faith.
- The museum was officially inaugurated by the Prime Minister of India in March 2026, coinciding with the auspicious occasion of Bhagwan Mahavir Jayanti.
- It serves as a massive repository of heritage, safely preserving over three lakh ancient manuscripts inscribed on palm leaves and birch bark.

Q 29. Ans. D

Explanation:

- In the IPL 2026 final, Royal Challengers Bengaluru (RCB) defeated the **Gujarat Titans** by 5 wickets to win their second consecutive IPL title.
- The championship match was played at the Narendra Modi Stadium in Ahmedabad on May 31, 2026.
- Batting first, the Gujarat Titans set a target of 156 (155/8). RCB successfully chased it down in just 18 overs (161/5), largely driven by Virat Kohli's unbeaten 75 off 42 balls.
- With this victory, RCB became only the third franchise in history, joining the Chennai Super Kings and Mumbai Indians, to successfully defend an IPL title.

Q 30. Ans. C

Explanation:

- The state-level agricultural mechanisation exhibition, '**Agro Bihar 2026**', was successfully organized at Gandhi Maidan in **Patna** from March 12 to 15, 2026.
- The mega exhibition was jointly organized by the Bihar Agriculture Department in collaboration with the Confederation of Indian Industry (CII).
- The core objective of the four-day event was to introduce farmers to modern farming techniques, AI-based agricultural technology, and provide a direct platform for top agricultural machinery manufacturers from across the country to showcase their innovations.
- **Additional Facts:**
 - ✓ **Subsidized Machinery:** During the event, the state government offered targeted subsidies on 91 different types of agricultural machinery to make them more accessible to local farmers, with special provisions for scheduled castes, scheduled tribes, and extremely backward classes.

- ✓ **Custom Hiring Centres:** To further support small and marginal farmers, the Bihar government emphasized the expansion of 'Custom Hiring Centres' across all districts. These centres allow farmers to rent expensive, heavy agricultural equipment rather than buying them outright.
- ✓ **Scale and Reach:** Recognized as the largest agricultural mechanization fair in eastern India, the 2026 edition saw the participation of over 35,000 farmers and featured more than 100 dedicated stalls spanning a massive 3.25 lakh square feet area.

Q 31. Ans. B

Explanation:

- **Auxins (a - 2):** Primarily responsible for cell elongation in shoots, regulating apical dominance (where the central stem grows more strongly than side stems), and initiating root formation.
- **Gibberellins (b - 3):** Play a crucial role in breaking seed dormancy, triggering seed germination, and promoting the elongation of the stem's internodes.
- **Cytokinins (c - 4):** Synthesized in regions of rapid cell division (such as root apices). They actively promote cell division (cytokinesis) and delay the aging of leaves (senescence), a process scientifically referred to as the Richmond-Lang effect.
- **Abscisic Acid (d - 1):** Often called the "stress hormone," it acts as a general inhibitor of plant growth. It promotes seed dormancy to ensure seeds only germinate under favorable conditions and triggers the closure of stomata during extreme water stress.
- **Additional Facts:**
 - ✓ **Ethylene (The Gaseous Hormone):** It is the fifth major class of plant hormones. Ethylene is highly unique because it is a gas at room temperature, primarily functioning to accelerate the ripening of fleshy fruits and promote the abscission (shedding) of leaves and flowers.
 - ✓ **Antagonistic Action:** Abscisic acid (ABA) and Gibberellins act in direct opposition to each other regarding seed germination. While Gibberellins break seed dormancy, ABA actively enforces it.
 - ✓ **Discovery of Gibberellins:** They were first identified in Japan by researchers studying 'bakanae' (foolish seedling) disease in rice. The disease, which caused unnatural and weak

stem elongation, was traced to the fungus *Gibberella fujikuroi*.

- ✓ **Synthetic Auxins:** Artificially synthesized auxins, such as 2,4-D (2,4-Dichlorophenoxyacetic acid), are widely used in modern agriculture as selective herbicides. They effectively kill dicotyledonous (broad-leaved) weeds without harming monocotyledonous cereal crops.

Q 32. Ans. A

Explanation:

- Temperature is directly proportional to the average **kinetic energy** of particles in a substance.
- When a substance is heated, the thermal energy is absorbed by the particles, causing them to move, vibrate, or rotate much faster—meaning their kinetic energy increases.
- In a liquid, once these particles gain sufficient kinetic energy, they are able to overcome the intermolecular forces of attraction binding them together. When this threshold is crossed, they break free and escape into the gaseous or vapour state.
- **Mass** remains constant regardless of temperature or state (Law of Conservation of Mass); **Density** typically decreases as a substance gets hotter and expands; and while intermolecular **potential energy** does change as the bonds break during the phase transition, it is the initial gain in kinetic energy that drives the rapid movement required to enter the vapour state.
- **Additional Facts:**
 - ✓ **Evaporation vs. Boiling:** Both involve gaining kinetic energy to become vapor, but they differ fundamentally. Evaporation is a surface phenomenon that occurs at any temperature below the boiling point (only surface particles escape). Boiling is a bulk phenomenon that happens at a specific temperature where particles throughout the entire liquid gain enough energy to form vapor bubbles.
 - ✓ **Latent Heat of Vaporisation:** This is the exact amount of heat energy required to change 1 kg of a liquid into a gas at atmospheric pressure at its boiling point.
 - ✓ **Sublimation:** In certain substances (like camphor, iodine, or solid carbon dioxide/dry ice), particles gain kinetic energy so rapidly upon heating that they bypass the liquid phase entirely, transitioning directly from a solid into a vapour state.

Q 33. Ans. D

Explanation:

- **Statement 4 is correct:** Fungi lack chlorophyll and cannot synthesize their own food. They are heterotrophic, specifically functioning mostly as saprophytes (or saprotrophs). They secrete digestive enzymes into the surrounding dead and decaying organic matter to break it down into simpler molecules, which they then absorb.
- **Statement 1 is incorrect:** Fungi are never autotrophic. Plants and some bacteria are autotrophic as they can produce their own food via photosynthesis or chemosynthesis.
- **Statement 2 is incorrect:** Unlike plants, which have cell walls made of cellulose, the cell walls of fungi are primarily composed of **chitin** (a complex structural polysaccharide) and glucans.
- **Statement 3 is incorrect:** Fungi have diverse reproductive strategies and can reproduce both asexually (via budding, fragmentation, or asexual spores like conidia) and sexually (via sexual spores like ascospores or basidiospores).
- **Additional Facts:**
 - ✓ **Reserve Food:** Much like animals, fungi store their reserve food energy primarily in the form of **glycogen** and oil globules, whereas plants store it as starch.
 - ✓ **Symbiotic Relationships:** Fungi frequently form mutually beneficial partnerships. **Lichens** are a symbiotic association between a fungus and an alga (or cyanobacterium). **Mycorrhizae** represent a symbiotic relationship between a fungus and the roots of higher vascular plants, aiding the plant in nutrient and water absorption.
 - ✓ **Chitin's Chemical Makeup:** The chitin found in fungal cell walls is also the exact same tough carbohydrate polymer that makes up the exoskeletons of arthropods (like insects, crabs, and spiders).
 - ✓ **Study of Fungi:** The scientific study of fungi is known as **Mycology**.

Q 34. Ans. A

Explanation:

- **Assertion (A) is true:** Genetically modified crops can be specifically engineered to resist pests. A prime example is Bt cotton, which produces its own internal toxin against the bollworm, thereby significantly reducing the farmer's need to spray external chemical pesticides.

- **Reason (R) is true and explains (A):** The fundamental mechanism that allows a GM crop to possess a novel trait like pest resistance (the Assertion) is exactly what is described in the Reason. Scientists artificially introduce specific foreign genes (such as the cry gene from a soil bacterium) into the plant's genome so that it expresses that desired trait.

Additional Facts:

- ✓ **Bt Cotton in India:** Introduced in 2002, Bt cotton remains the **only** genetically modified crop legally approved for commercial cultivation in India.
- ✓ **Regulatory Authority:** The apex body responsible for the appraisal and approval of activities involving large-scale use of hazardous microorganisms and recombinants, including the commercial release of GM crops in India, is the **Genetic Engineering Appraisal Committee (GEAC)**. It operates under the Ministry of Environment, Forest and Climate Change (MoEFCC).
- ✓ **Golden Rice:** This is a prominent example of biofortification via genetic modification. Golden rice has been genetically engineered to biosynthesize beta-carotene (a precursor to Vitamin A) in the edible parts of rice, aiming to combat severe Vitamin A deficiency in developing nations.
- ✓ **Mustard DMH-11:** Dhara Mustard Hybrid-11 (DMH-11) is a genetically modified mustard variety developed by Delhi University researchers. It has been a subject of intense debate and regulatory scrutiny in India regarding its environmental release for seed production and testing.

Q 35. Ans. A

Explanation:

- **Deposition** (also known as desublimation) is a phase transition where a substance transitions directly from a gas to a solid, completely bypassing the intermediate liquid phase.
- It occurs when thermal energy is rapidly removed from a gas, causing its particles to lose kinetic energy and lock directly into a rigid solid structure.
- Because it involves the release of heat energy to the surroundings, deposition is an **exothermic** process.
- **Additional Facts:**
 - ✓ **The Opposite Process:** The reverse of deposition is **sublimation**, where a solid transitions directly into a gas without

becoming a liquid (e.g., dry ice turning into carbon dioxide gas).

- ✓ **Frost Formation:** A common, everyday example of deposition is the formation of frost on a cold winter morning. Water vapor in the sub-freezing air directly deposits as solid ice crystals on windows or grass, without first condensing into liquid water droplets.
- ✓ **Snow in Clouds:** Snowflakes form in clouds through deposition. Water vapor in the high, freezing altitudes directly crystallizes into ice on dust particles.
- ✓ **Vapour Deposition:** In modern manufacturing (like making semiconductors or synthetic diamonds), a technique called Chemical Vapour Deposition (CVD) is used to deposit a solid, high-purity, high-performance film directly from a gaseous state onto a substrate.

Q 36. Ans. D

Explanation:

- **Apoptosis** is the highly regulated and genetically programmed process of cell death. It is a controlled, predictable routine where a cell actively participates in its own destruction at the cellular and molecular level.
- This process is vital for maintaining the health of the organism, allowing for the safe disposal of old, unnecessary, or damaged cells without harming the surrounding tissue.
- **Why the others are incorrect:**
 - ✓ **Necrosis** is the unprogrammed, accidental death of cells caused by external factors such as injury, infection, or toxins. Unlike apoptosis, necrosis causes inflammation and damage to surrounding tissues.
 - ✓ **Ageing** (senescence) is the gradual deterioration of functional characteristics over time, not a specific cellular self-destruction protocol.
 - ✓ **Degeneration** refers to the general structural or functional decline of tissues, often as a result of disease or aging.
- **Additional Facts:**
 - ✓ **Embryonic Development:** Apoptosis plays a crucial role in shaping physical features during embryonic development. For example, the separation of human fingers and toes occurs because the cells forming the webbing between them undergo apoptosis.
 - ✓ **Cancer Connection:** A failure in the apoptotic process is a primary hallmark of cancer. When cells with damaged DNA refuse to undergo

programmed cell death, they continue to multiply uncontrollably, forming tumors.

- ✓ **Executioner Enzymes:** The process of apoptosis is driven by a specialized family of protease enzymes called **caspases**. Once activated, these enzymes systematically dismantle the cell's structural proteins and DNA.
- ✓ **Phagocytosis:** Once a cell has undergone apoptosis, it breaks apart into smaller, membrane-bound fragments called "apoptotic bodies." These are quickly recognized and engulfed by immune cells (macrophages) in a process called phagocytosis, leaving no trace and preventing inflammation.

Q 37. Ans. B

Explanation:

- In aerated drinks like soda water, **carbon dioxide** is the gas dissolved as the solute, while water acts as the solvent.
- Carbon dioxide is forced into the liquid under high pressure. When the bottle is opened, the pressure drops abruptly, causing the gas to rapidly escape in the form of bubbles, a process known as effervescence.
- The dissolved carbon dioxide reacts slightly with the water to form a weak acid called carbonic acid, which gives soda its characteristic tangy, sharp taste.
- **Additional Facts:**
 - ✓ **Henry's Law:** The science behind carbonating beverages is governed by Henry's Law, which states that the solubility of a gas in a liquid is directly proportional to the pressure of that gas above the liquid. This is exactly why soda cans are highly pressurised before sealing.
 - ✓ **Temperature and Gas Solubility:** Unlike most solid solutes (like sugar or salt), the solubility of gases in liquids decreases as temperature rises. This is why warm soda goes flat much faster than cold soda when left exposed to the air.
 - ✓ **Universal Solvent:** Water is often referred to as the "universal solvent" because it is capable of dissolving more substances than any other liquid, making it the base for almost all commercial beverages.
 - ✓ **Dry Ice:** The solid form of carbon dioxide is known as dry ice. At normal atmospheric pressure, it undergoes sublimation (turning directly from a solid to a gas) at -78.5 degrees Celsius, bypassing the liquid phase entirely.

Q 38. Ans. A

Explanation:

- **Statement I is correct:** By definition, organisms that can synthesize their own food (carbohydrates) from simple inorganic substances using light energy and chlorophyll are autotrophs. Since all green plants possess chlorophyll and perform photosynthesis, they are fundamentally autotrophic.
- **Statement II is correct:** Insectivorous plants, such as the pitcher plant, are green and fully capable of photosynthesis to meet their carbon and energy needs. However, they are considered partially heterotrophic because they typically grow in soils that are severely deficient in certain essential nutrients, most notably nitrogen. They evolved mechanisms to trap and digest insects to supplement this specific nutritional deficit.
- **Additional Facts:**
 - ✓ **Nitrogen Deficiency:** The primary evolutionary driver for insectivorous (carnivorous) plants is the lack of available nitrogen in their native habitats, which are often acidic bogs or rocky outcrops. They use enzymes to break down the proteins in trapped insects to extract nitrogen.
 - ✓ **Other Insectivorous Plants:** Besides the pitcher plant (*Nepenthes*), other common examples include the Venus flytrap (*Dionaea muscipula*), Sundew (*Drosera*), and Bladderwort (*Utricularia*).
 - ✓ **Complete Parasites:** Not all plants are autotrophic. Some plants, like *Cuscuta* (dodder or Amarbel), have lost their chlorophyll entirely through evolution. They are completely heterotrophic and rely entirely on a host plant for both water and food, using special root-like structures called haustoria to penetrate the host's vascular system.
 - ✓ **Saprotrophic Plants:** Certain plants, like the Indian pipe (*Monotropa*), also lack chlorophyll and are entirely white or pale. They obtain their nutrients from dead and decaying organic matter in the soil, usually via a symbiotic relationship with fungi (mycoheterotrophy).

Q 39. Ans. C

Explanation:

- The **nephron** is the microscopic structural and functional unit of the kidney.
- It operates in two main stages to ensure efficiency: first, it filters the blood under high pressure to remove

water, urea, salts, and glucose (ultrafiltration). Second, as this filtrate travels through the tubule, it intelligently reabsorbs the essential substances the body still needs (like glucose, amino acids, and most of the water) back into the bloodstream, leaving only the concentrated waste as urine.

- **The other options describe the functions of different organs:** the urinary bladder stores urine, the large intestine removes solid digestive waste, and the alveoli in the lungs handle gas exchange.
- **Additional Facts:**
 - ✓ **Massive Scale:** Each human kidney contains approximately **1 to 1.5 million** nephrons. Despite their microscopic size, they collectively filter about 180 liters of blood plasma every single day.
 - ✓ **Bowman's Capsule:** The actual filtration of blood begins in a cup-like sac at the start of the nephron called Bowman's capsule, which encloses a cluster of tiny, porous blood vessels known as the glomerulus.
 - ✓ **Selective Reabsorption:** The process is highly selective. For instance, in a healthy individual, 100% of the filtered glucose is reabsorbed. The presence of glucose in the final urine is often a primary indicator of diabetes.
 - ✓ **Dialysis (Artificial Kidney):** When nephrons are severely damaged and the kidneys fail, the blood must be filtered artificially using a machine, a life-saving medical procedure known as hemodialysis.

Q 40. Ans. D

Explanation:

- **Diclofenac**, a common non-steroidal anti-inflammatory drug (NSAID), is the primary cause of the catastrophic decline in vulture populations across the Indian subcontinent.
- The drug was widely used by farmers as a painkiller and to treat inflammation in cattle and other livestock.
- When treated livestock died and were left in the open, vultures fed on their carcasses. Vultures lack the specific enzymes needed to break down diclofenac. Even trace amounts of the drug in the meat caused them to suffer from acute kidney failure and visceral gout (accumulation of uric acid in organs), leading to death within a few days.
- **Additional Facts:**
 - ✓ **Population Crash:** The use of veterinary diclofenac led to a devastating 97% to 99% decline in the populations of three distinct

Gyps vulture species in India (the White-rumped Vulture, the Indian Vulture, and the Slender-billed Vulture) during the 1990s and early 2000s.

- ✓ **Government Ban:** Recognizing the ecological disaster, the Government of India officially banned the veterinary use of diclofenac in **2006**, though illegal use of the human formulation for cattle persisted for some years.
- ✓ **Safe Alternatives:** **Meloxicam** and **Tolfenamic acid** are other NSAIDs that have been scientifically tested and proven to be safe for vultures, and are currently promoted as alternatives to diclofenac for treating livestock.
- ✓ **Ecological Impact:** The near-extinction of vultures led to a massive increase in the population of feral dogs and rats that fed on rotting carcasses, which subsequently caused a spike in zoonotic diseases like rabies.

Q 41. Ans. B

Explanation:

- **Statement 1 is correctly matched:** Suckling in newborn mammals is a classic example of an **instinct**, a complex, innate (inborn) pattern of behavior that is biologically hardwired and does not need to be taught.
- **Statement 2 is correctly matched:** **Migration** is the seasonal, long-distance movement of animals (like birds, whales, or monarch butterflies) to find favorable climates, better food sources, or safe breeding grounds.
- **Statement 3 is incorrectly matched:** A dog sitting on command after training is a **conditioned response** (a type of learned behavior), not a reflex. A **reflex** is an involuntary, rapid, and automatic biological response to a stimulus, such as a dog shaking off water or a human quickly pulling a hand away from a hot surface without thinking.
- **Statement 4 is correctly matched:** A chimpanzee using a stick to extract termites is a **learned behavior** (specifically, cognitive learning and tool use). It is acquired through experience, problem-solving, and observing other members of the troop.
- **Additional Facts:**
 - ✓ **Classical Conditioning:** The most famous study of conditioned learned behavior was conducted by Russian physiologist **Ivan Pavlov**. He famously trained dogs to involuntarily salivate at the sound of a bell by repeatedly associating the sound with the presentation of food.

- ✓ **Ethology:** The scientific study of animal behavior, especially as it occurs in their natural environments, is known as **ethology**.
- ✓ **Imprinting:** A highly specialised phase of learning that occurs during a critical early period in an animal's life. The most famous example was demonstrated by scientist Konrad Lorenz, who showed that newly hatched geese would "imprint" on the first moving object they saw (which was Lorenz himself) and follow him around as if he were their mother.
- ✓ **Habituation:** This is considered the simplest form of animal learning. It occurs when an animal learns to completely ignore a repeated, harmless stimulus over time (for example, crows eventually learn to ignore a scarecrow in a field).

Q 42. Ans. B

Explanation:

- In a healthy individual, the pulse rate measured at the wrist (or any other pulse point) is **equal** to the heart rate.
- Every time the left ventricle of the heart contracts to pump blood into the aorta, it generates a wave of high pressure that travels through the entire arterial system.
- This expanding and recoiling of the arteries is what feels like a "pulse." Therefore, one heartbeat produces exactly one pulse wave.
- **Additional Facts:**
 - ✓ **Normal Range:** For a healthy human adult at rest, the normal heart rate (and thus pulse rate) ranges between **60 and 100 beats per minute (bpm)**.
 - ✓ **Athletic Bradycardia:** Highly trained athletes often have resting heart rates well below 60 bpm (sometimes in the 40s). This is not a sign of illness; rather, their heart muscles are so strong and efficient that they can pump the required volume of blood with fewer beats.
 - ✓ **Common Pulse Points:** While the **radial artery** at the wrist is the most common place to check a pulse, it can also be easily felt at the **carotid artery** (in the neck), the brachial artery (inside the elbow), and the femoral artery (in the groin).
 - ✓ **Pulse Deficit:** In certain medical conditions like atrial fibrillation (an irregular heartbeat), the heart may beat without pumping enough blood to create a palpable pressure wave in the

wrist. In this specific unhealthy scenario, the radial pulse rate can actually be lower than the true heart rate (a phenomenon called a “pulse deficit”).

Q 43. Ans. B

Explanation:

- **Statement 1 is correct:** Carbon fibre is renowned for its extremely high tensile strength, meaning it can withstand immense pulling or stretching forces without breaking.
- **Statement 2 is incorrect:** Carbon fibre is actually a good conductor of heat (and electricity). In fact, certain types of pitch-based carbon fibres have a thermal conductivity that surpasses even that of copper.
- **Statement 3 is correct:** It is highly chemically inert and resistant to corrosion. Unlike steel, carbon fibre will not rust when exposed to moisture or most chemicals.
- **Statement 4 is correct:** Its most valuable property is its exceptional strength-to-weight ratio. It is significantly lighter than steel or aluminium while offering comparable or superior strength, making it ideal for high-performance applications.
- **Additional Facts:**
 - ✓ **Precursor Materials:** About 90% of all carbon fibres are manufactured from a polymer called **Polyacrylonitrile (PAN)**. The remaining 10% are made from rayon or petroleum pitch.
 - ✓ **Electrical Conductivity:** Because carbon fibre conducts electricity well, it must be carefully insulated when used near electronic components to prevent short circuits. Additionally, if it is placed in direct contact with certain metals (like aluminium) in a moist environment, it can cause severe galvanic corrosion of the metal.
 - ✓ **Anisotropy:** The mechanical and thermal properties of carbon fibre are highly **anisotropic**, meaning they are directional. The fibre is incredibly strong and conductive along its longitudinal axis (the direction the fibres are pulled), but much weaker across its width.
 - ✓ **Carbon-Carbon Composites:** When carbon fibres are embedded within a carbon matrix, the resulting material (carbon-carbon composite) can withstand extreme temperatures (up to nearly 3000°C) without losing its strength, making it crucial for spacecraft heat shields and Formula One brake discs.

Q 44. Ans. A

Explanation:

- **Statement 1 is incorrect:** A colloidal solution appears to be homogeneous, but it is actually a **heterogeneous** mixture. For example, under a microscope, milk clearly shows distinct fat droplets suspended in liquid.
- **Statement 2 is correct:** Colloidal particles are typically between 1 and 1000 nanometers in diameter, which is far too small to be seen individually with the naked eye.
- **Statement 3 is correct:** Colloids are quite stable. Unlike suspensions (like sand in water), the particles in a colloid will not settle to the bottom under the influence of gravity if left undisturbed.
- **Statement 4 is incorrect:** The particles of a colloid are small enough to easily pass through the pores of ordinary filter paper, meaning they **cannot** be separated by standard filtration. A technique like centrifugation is required.
- **Statement 5 is correct:** Colloidal particles are large enough to scatter a beam of visible light passing through them, a phenomenon that makes the path of the light visible.
- **Additional Facts:**
 - ✓ **Tyndall Effect:** The phenomenon described in Statement 5, where colloidal particles scatter a beam of light, is called the Tyndall effect. This is the same effect seen when a fine beam of sunlight enters a dust-filled room.
 - ✓ **Brownian Motion:** Colloidal particles are constantly bombarded by the molecules of the medium they are suspended in. This causes the particles to move in a continuous, random, zig-zag path known as Brownian motion, which is the primary reason the particles do not settle.
 - ✓ **Two Components:** Every colloid consists of two phases: the **dispersed phase** (the solute-like particles that are suspended) and the **dispersion medium** (the solvent-like substance in which the particles are suspended).
 - ✓ **Types of Colloids:** Colloids are classified based on the state of their components. For instance, an **aerosol** is a liquid or solid dispersed in a gas (like fog or smoke), while an **emulsion** is a liquid dispersed in another liquid (like milk or mayonnaise).

Q 45. Ans. B

Explanation:

- **Alveoli:** These are the microscopic, balloon-like air sacs located at the very end of the respiratory tree.

Their extremely thin walls and dense covering of blood capillaries make them the **primary functional site where oxygen diffuses into the blood and carbon dioxide diffuses out.**

- **Bronchi:** These are the major air passages that branch off from the trachea into the lungs. They serve purely as **conducting tubes** to transport air deeper into the respiratory system, not as sites for gas exchange.
- **Trachea:** Also known as the **windpipe**, this is the main cartilage-ringed tube connecting the larynx to the bronchi. It keeps the main airway open and filters incoming air, but it does not participate in gas exchange.
- **Pulmonary artery:** This major blood vessel carries deoxygenated blood from the right ventricle of the heart to the lungs. While it transports the **blood to the lungs**, the actual gas exchange happens at the capillary-alveolar interface, not within the artery itself.
- **Additional Facts:**
 - ✓ **Massive Surface Area:** If all the hundreds of millions of alveoli in an average human's lungs are flattened, their combined surface area would be roughly the size of a tennis court (about 50 to 75 square meters). This large area is crucial for meeting the body's rapid oxygen demands.
 - ✓ **Surfactant:** The inner lining of the alveoli is coated with a fluid called pulmonary surfactant (produced by specialized Type II alveolar cells). This substance lowers surface tension, preventing the delicate air sacs from collapsing on themselves when exhaled.
 - ✓ **Diffusion Gradient:** The actual exchange of gases is a passive process driven by partial pressure differences. Oxygen naturally moves from the high-concentration air in the alveoli to the lower-concentration blood, while carbon dioxide moves in the exact opposite direction.
 - ✓ **Hemoglobin's Role:** Once oxygen crosses the alveolar membrane into the blood, roughly 98% of it immediately binds to hemoglobin proteins inside red blood cells to be efficiently transported to tissues throughout the entire body.

Q 46. Ans. A

Explanation:

- **Milk:** Milk is an emulsion, a type of colloid where fat globules and proteins are dispersed in water. Because its particles are relatively large (between 1 and 1000 nanometers), it successfully scatters a

beam of light passing through it, exhibiting the Tyndall effect.

- **Sugar solution:** This is a true solution. The sugar molecules dissolve completely and homogeneously into individual molecules that are too small to scatter visible light, meaning it does not exhibit the Tyndall effect.
- **Starch solution:** When starch is mixed with water, it forms a macromolecular colloid rather than a true solution. The large starch molecules remain suspended and are large enough to scatter light, exhibiting the Tyndall effect.
- **Sodium chloride solution:** Similar to sugar water, this is a true solution (saltwater). The completely dissolved sodium and chloride ions are microscopic and cannot scatter a light beam.
- **Additional Facts:**
 - ✓ **The Tyndall Effect in Nature:** This effect is seen in everyday life when sunlight shines through a dust-filled room, or when car headlights pass through fog. The microscopic dust particles or water droplets in the air (which act as aerosols, a type of colloid) scatter the light beams, making them visible.
 - ✓ **John Tyndall:** The phenomenon is named after the 19th-century Irish physicist John Tyndall, who first extensively studied and described the light-scattering properties of suspended particles.
 - ✓ **Blue Eyes:** The Tyndall effect also explains the appearance of blue eyes in humans. There is no actual blue pigment in the iris; instead, translucent layers in the iris scatter short-wavelength blue light much like particles in a colloid, reflecting it back to the observer.
 - ✓ **Rayleigh vs. Tyndall:** While both involve light scattering, they are distinct. Rayleigh scattering involves particles much smaller than the wavelength of light (like individual oxygen and nitrogen molecules, which is why the sky is blue). Tyndall scattering involves slightly larger particles, typical of colloidal sizes.

Q 47. Ans. D

Explanation:

- **Statement A is correct:** The primary respiratory control center, which dictates the basic, automatic rhythm of breathing, is the respiratory rhythm center located in the medulla oblongata at the base of the brain.
- **Statement B is correct:** A rise in carbon dioxide in the blood lowers blood pH (making it more acidic).

Specialized chemoreceptors detect this chemical change and immediately signal the medulla to increase the breathing rate to expel the excess carbon dioxide. In fact, carbon dioxide buildup is the primary driving force behind the urge to breathe, not a lack of oxygen.

- **Statement C is correct:** While carbon dioxide is the primary driver, a significant drop in blood oxygen levels can also trigger peripheral chemoreceptors to stimulate breathing, acting as a crucial secondary backup system (especially at high altitudes).
- **Statement D is incorrect:** Breathing is entirely dependent on neural signals generated in the brain. The respiratory muscles (primarily the diaphragm and intercostal muscles) are skeletal muscles that will not contract without specific nerve impulses traveling from the brainstem down the spinal cord via the phrenic and intercostal nerves.
- **Additional Facts:**
 - ✓ **The Pons:** The pons, another crucial part of the brainstem located just above the medulla, contains the pneumotaxic center. It acts as a modulator, essentially providing an “off switch” to limit inspiration. This prevents the lungs from over-inflating and helps fine-tune the breathing rate.
 - ✓ **Voluntary Override:** While breathing is primarily an involuntary, automatic process driven by the brainstem, we possess temporary voluntary control via the cerebral cortex (allowing us to speak, sing, or hold our breath underwater).
 - ✓ **Breath-Holding Limit:** Prolonged voluntary breath-holding (apnea) leads to an accumulation of carbon dioxide in the blood. This rising concentration eventually generates a chemical stimulus potent enough to override conscious cortical control, triggering an involuntary inhalation. In the event that breath-holding results in syncope (fainting), the autonomic control centers in the brainstem automatically reinstate the respiratory cycle.
 - ✓ **Chemoreceptor Locations:** Peripheral chemoreceptors are strategically placed in the carotid bodies (monitoring blood flowing to the brain) and the aortic bodies (monitoring blood just leaving the heart) to continuously sample the blood’s chemical composition.

Q 48. Ans. B

Explanation:

- In the context of dietary minerals, **Iron** is classified

as a micronutrient (specifically a trace mineral), not a macronutrient or macromineral.

- The human body requires trace minerals in very small amounts (typically less than 100 milligrams per day).
- **Calcium, Phosphorus, and Potassium** are classified as macrominerals (major minerals). The body requires these in significantly larger quantities (hundreds to thousands of milligrams per day) to maintain bone density, fluid balance, and proper nerve and muscle function.
- **Additional Facts:**
 - ✓ **The “Macro” Definition:** In broader nutritional science, the word “macronutrient” most commonly refers to **carbohydrates, proteins, and fats**, the primary nutrient classes that provide the body with caloric energy. Minerals and vitamins are collectively known as micronutrients, though minerals are further subdivided by quantity into major and trace.
 - ✓ **Iron’s Crucial Role:** Despite being needed in tiny amounts, iron is indispensable. It is the central component of **hemoglobin**, the protein in red blood cells responsible for binding oxygen in the lungs and transporting it to every tissue in the body.
 - ✓ **Other Trace Minerals:** Alongside iron, other essential trace minerals that the body only needs in microscopic amounts include zinc, copper, iodine, selenium, and fluoride.
 - ✓ **The Bone Builders:** Calcium and phosphorus are the two most abundant minerals in the human body. They work closely together, combining to form hydroxyapatite, the hard crystal matrix that gives skeletal bones and teeth their rigid, durable structure.

Q 49. Ans. D

Explanation:

- **Statement 1 is correct:** Skeletal muscle fibers possess a distinct striated (striped) appearance under microscopic examination. This is due to the highly organized, alternating arrangement of the contractile proteins actin and myosin into repeating units called sarcomeres.
- **Statement 2 is correct:** Skeletal muscles function under voluntary control. Their contraction is initiated by conscious neural signals transmitted via the somatic nervous system.
- **Statement 3 is correct:** Skeletal muscle cells (fibers) are uniquely long, cylindrical, and multinucleate.

During embryonic development, numerous precursor cells (myoblasts) fuse to form a single continuous fiber, resulting in multiple nuclei sharing a common cytoplasm.

- **Statement 4 is incorrect:** Skeletal muscles do not contract without conscious control. Involuntary contraction is a characteristic of cardiac and smooth muscle tissues, which are governed by the autonomic nervous system, whereas skeletal muscles require somatic innervation.

- **Additional Facts:**

- ✓ **Other Muscle Types:** The human body has three types of muscle tissue. **Cardiac muscle** (found only in the heart) is striated but involuntary. **Smooth muscle** (found lining the digestive tract, blood vessels, and internal organs) is non-striated and involuntary.
- ✓ **The Sarcomere:** The fundamental, functional unit of contraction within a skeletal muscle is the sarcomere. It is the microscopic segment between two specific protein boundaries (Z-lines) that physically shortens when the actin and myosin filaments slide past one another.
- ✓ **Fatigue:** Unlike cardiac muscle, which beats continuously for a lifetime without tiring, skeletal muscles are highly susceptible to fatigue. During intense exercise, when oxygen delivery cannot keep up with energy demand, they produce lactic acid as a byproduct of anaerobic respiration.
- ✓ **Muscle Attachment:** Skeletal muscles are almost always attached to the bones of the skeleton by strong, inelastic connective tissue called **tendons** (whereas ligaments connect bone to bone).

Q 50. Ans. A

Explanation:

- **Sodium silicate and sodium sulphate:** Sodium sulphate is extensively utilised as a filler in synthetic powdered detergents to increase bulk and ensure the powder remains free-flowing without clumping. Sodium silicate serves as a builder and corrosion inhibitor, which protects internal washing machine components from rusting and assists in keeping suspended dirt from redepositing onto fabrics.
- **Sodium chloride and calcium carbonate:** While sodium chloride is occasionally used as a viscosity modifier (thickener) in some liquid formulations, neither of these compounds constitutes the primary

bulk or builder ingredients for manufacturing standard synthetic washing powders.

- **Ammonium sulphate and urea:** These nitrogen-rich compounds are primarily synthesised for use in agricultural fertilisers and do not function as cleaning agents or builders in detergents.
- **Citric acid and sodium bicarbonate:** Although these chemicals are common in specialized, environmentally friendly cleaning alternatives or effervescent products, they are not the foundational chemicals utilised in the large-scale industrial manufacture of synthetic washing powders.

- **Additional Facts:**

- ✓ **Surfactants:** The primary active cleaning agents in synthetic detergents are surface-active agents (surfactants), such as sodium alkylbenzene sulfonates. These molecules possess a hydrophilic (water-attracting) head and a hydrophobic (water-repelling) tail, enabling them to surround and lift oil and grease from fabrics.
- ✓ **Hard Water Compatibility:** A significant advantage of synthetic detergents over traditional soaps is their efficacy in hard water. Detergents do not react with calcium and magnesium ions present in hard water to form insoluble precipitates (scum), thereby retaining their full cleaning capacity.
- ✓ **Optical Brighteners:** Commercial washing powders frequently incorporate fluorescent whitening agents, known as optical brighteners. These compounds absorb invisible ultraviolet light and re-emit it as visible blue light, which neutralizes the natural yellowish tint of aging fabrics and creates the illusion of enhanced whiteness.
- ✓ **Enzymatic Additives:** Modern detergent formulations often include specific biological enzymes, such as proteases, lipases, and amylases. These enzymes act as biological catalysts to efficiently break down complex protein, fat, and carbohydrate-based stains at lower wash temperatures.

Q 51. Ans. C

Explanation:

- **Statement 1 is correct:** The primary physiological function of the large intestine is the absorption of water and electrolytes (such as sodium and chloride) from the remaining indigestible food matter, thereby transforming liquid chyme into semi-solid feces.

- **Statement 2 is incorrect:** The large intestine does not secrete digestive enzymes, nor does it participate in the chemical digestion of proteins. The enzymatic breakdown of proteins, carbohydrates, and lipids is completed entirely within the stomach and the small intestine.
- **Statement 3 is correct:** The large intestine hosts a vast microbiome of commensal bacteria. These microorganisms ferment indigestible carbohydrates and synthesize several essential vitamins, most notably vitamin K and various B vitamins, which are subsequently absorbed through the intestinal mucosa into the bloodstream.
- **Additional Facts:**
 - ✓ **Anatomical Segments:** The large intestine is anatomically subdivided into four main regions: the cecum, the colon (further divided into ascending, transverse, descending, and sigmoid sections), the rectum, and the anal canal.
 - ✓ **Vermiform Appendix:** Attached to the cecum is the appendix, a blind-ended tube containing dense lymphoid tissue. While historically considered a vestigial structure, contemporary research suggests it may function as a safe reservoir for beneficial gut bacteria to facilitate repopulation following gastrointestinal illnesses.
 - ✓ **Haustra:** The longitudinal smooth muscle of the colon, known as the taenia coli, is shorter than the colon itself. This structural arrangement causes the intestinal wall to pucker into small, pouch-like sacs called haustra, imparting a characteristic segmented appearance.
 - ✓ **Goblet Cells:** The mucosal epithelium of the large intestine is highly populated with goblet cells. These specialized cells secrete abundant mucus, which serves to lubricate the intestinal lumen and facilitate the smooth passage of progressively dehydrating fecal matter.

Q 52. Ans. A

Explanation:

- **Statement 1 is correct:** Ethanethiol, formally known as ethyl mercaptan, is a volatile, colorless, and highly flammable organosulfur compound distinguished by an extremely pungent, offensive odor.
- **Statement 2 is correct:** Liquefied Petroleum Gas (LPG) is naturally odorless. Ethanethiol is

intentionally infused into commercial LPG as an odorant. This serves as a critical safety mechanism, enabling the olfactory detection of gas leaks prior to the accumulation of combustible or explosive concentrations.

- **Statement 3 is incorrect:** The characteristic foul odor of ethanethiol originates from the presence of a sulfhydryl or thiol functional group (-SH), rather than a hydroxyl group (-OH). The hydroxyl functional group is characteristic of alcohols (such as ethanol), which possess entirely distinct chemical and olfactory properties.
- **Additional Facts:**
 - ✓ **Chemical Structure:** The molecular formula for ethanethiol is $\text{CH}_3\text{CH}_2\text{SH}$. It is structurally analogous to ethanol ($\text{CH}_3\text{CH}_2\text{OH}$), with a sulfur atom directly substituting the oxygen atom within the functional group.
 - ✓ **Olfactory Threshold:** The human olfactory system exhibits extraordinary sensitivity to thiol compounds. Ethanethiol can be detected by human receptors at atmospheric concentrations as minimal as one part per 2.8 billion.
 - ✓ **Related Compounds:** Methanethiol (methyl mercaptan) is a related sulfur compound that occurs naturally as a byproduct of microbial metabolism in decaying organic matter, and it is a primary contributor to the odors of human halitosis and flatulence.
 - ✓ **Toxicity:** Although ethanethiol is toxic at elevated concentrations, the trace quantities utilized as a warning odorant in commercial LPG are physiologically insignificant and do not pose a toxicological hazard during a standard gas leak detection event.

Q 53. Ans. C

Explanation:

- **Centrifugal force:** During the spin cycle, the rapid rotation of the drum creates a substantial outward radial force (centrifugal force) relative to the rotating reference frame. This apparent force pushes the wet clothing against the perforated drum wall. Because the water is not physically constrained by the fabric, it continues its outward trajectory through the perforations, effectively separating from the clothes.
- **Gravitational force:** Although gravity exerts a continuous downward pull on all mass, it is entirely insufficient to overcome the surface tension and capillary action holding water within the microscopic pores of the fabric.

- **Electrostatic force:** This force governs the interactions between electrically charged particles. It plays no mechanical role in the bulk separation of water from textiles during a washing cycle.
- **Magnetic force:** Magnetic forces dictate interactions between magnetic materials or moving electrical charges and do not contribute to the mechanical extraction of water.
- **Additional Facts:**
 - ✓ **Inertial Frame Perspective:** From the perspective of classical Newtonian mechanics (an inertial reference frame), “centrifugal force” is considered a fictitious force. The drum provides the necessary centripetal force to keep the clothes moving in a circular path. The water droplets, lacking sufficient centripetal force acting upon them, travel in a straight tangential line due to inertia, escaping through the drum’s perforations.
 - ✓ **Laboratory Centrifuges:** The identical physical principle is utilized extensively in biochemical laboratories. High-speed centrifuges rapidly rotate test tubes to separate suspensions based on density, such as isolating red blood cells from blood plasma or separating isotopes.
 - ✓ **Capillary Adhesion:** Water adheres strongly to the hydrophilic fibers of clothing due to capillary action and molecular adhesion. The mechanical force exerted during the spin cycle must cross a critical threshold to overcome these molecular bonds.
 - ✓ **Relative Centrifugal Force (RCF):** Modern washing machines operate at spin speeds frequently exceeding 1,000 revolutions per minute (RPM). The resulting Relative Centrifugal Force generated at the drum’s perimeter can reach several hundred times the standard acceleration of terrestrial gravity.

Q 54. Ans. C

Explanation:

- **Modern physics recognizes exactly four fundamental interactions that govern the behavior of all matter and energy in the universe:** the gravitational force, the electromagnetic force, the strong nuclear force, and the weak nuclear force.
- **Additional Facts:**
 - ✓ **Strong Nuclear Force:** This is the most powerful of the four fundamental interactions. It is responsible for binding quarks together to form protons and neutrons, and for

holding these nucleons together within the atomic nucleus, overcoming the intense electromagnetic repulsion between positively charged protons. Its operational range is strictly subatomic.

- ✓ **Electromagnetic Force:** This interaction dictates the behavior of electrically charged particles. It possesses an infinite range and is responsible for virtually all macroscopic phenomena experienced outside of gravity, including friction, elasticity, light (electromagnetic radiation), and the chemical bonding of atoms.
- ✓ **Weak Nuclear Force:** This interaction governs specific forms of radioactive decay (such as beta decay) and plays a critical role in the nuclear fusion processes that power stars. It operates at an even shorter subatomic range than the strong nuclear force.
- ✓ **Gravity:** Despite being the most macroscopically apparent force and possessing an infinite range, gravity is exponentially the weakest of the four interactions at the microscopic scale. It currently remains the only fundamental force not successfully described by quantum mechanics and is excluded from the Standard Model of particle physics.

Q 55. Ans. B

Explanation:

- The escape velocity at the surface of the Earth is approximately **11.2 km/s** (or roughly **40,270 km/h**). This represents the minimum kinetic energy threshold required for a ballistic unpowered projectile to indefinitely escape Earth’s gravitational influence.
- **Additional Facts:**
 - ✓ **Mathematical Formula:** Escape velocity (v_e) is calculated using the equation $v_e = \sqrt{2GM/R}$, where G represents the universal gravitational constant, M is the mass of the planetary body, and R is its radius. It is mathematically independent of the mass of the escaping object.
 - ✓ In an idealised theoretical model devoid of atmospheric resistance, escape velocity functions as a scalar quantity. An object traveling at or above this threshold will escape the gravitational well regardless of its specific launch trajectory, provided the path does not intersect the planetary surface.

- ✓ Practical aerospace vehicles do not accelerate to 11.2 km/s instantaneously at the terrestrial surface, as the resulting atmospheric friction would induce extreme thermal destruction. Instead, launch vehicles utilise continuous chemical propulsion to gain altitude gradually, achieving escape velocity within the exosphere where aerodynamic drag is negligible.

Q 56. Ans. C

Explanation:

- **Statement 1 is correct:** Pressure is a scalar quantity. It is defined as the magnitude of the normal force applied per unit area ($P = F/A$). Because it represents the isotropic (equal in all directions) outward push within a fluid or the pure magnitude of force on a surface, it lacks a specific directional vector component in classical physics.
- **Statement 2 is incorrect:** Momentum is a vector quantity. It is mathematically defined as the product of an object's mass (a scalar) and its velocity (a vector). Because velocity inherently possesses direction, momentum must also possess both magnitude and direction ($p = mv$).
- **Statement 3 is correct:** Energy is a scalar quantity. Whether it is kinetic, potential, thermal, or chemical, energy represents the capacity of a physical system to perform work. It is measured purely in magnitude (Joules) without any associated spatial direction.
- **Statement 4 is correct:** Work is a scalar quantity. In physics, work is defined as the dot product of two vector quantities: force and displacement ($W = F \cdot d \cdot \cos(\theta)$). In vector mathematics, calculating the dot product of two vectors invariably yields a scalar result.
- **Additional Facts:**
 - ✓ While pressure is treated as a scalar in basic fluid mechanics, in advanced solid mechanics, stress (a related concept) is defined as a second-order tensor, as forces can be applied tangentially (shear stress) as well as normally.
 - ✓ Vectors cannot be added using simple arithmetic unless they are perfectly collinear. Instead, vector addition requires geometric rules, such as the parallelogram law or the head-to-tail method, to accurately calculate the resultant magnitude and direction.
 - ✓ While the dot product of two vectors (like force and displacement) yields a scalar (work), the cross product of two vectors yields a new, third vector that is perpendicular to the plane

containing the original two. For example, torque is a vector quantity derived from the cross product of a radius vector and a force vector.

Q 57. Ans. C

Explanation:

- **Doppler effect:** In astrophysics, the observational evidence for the expansion of the universe is rooted in the **Doppler effect**, specifically observed as a "redshift." When light from distant galaxies is analysed, its spectral lines are shifted toward the longer-wavelength (red) end of the electromagnetic spectrum, indicating that these galaxies are receding from the observer.
- **Zeeman effect:** The Zeeman effect describes the splitting of atomic spectral lines in the presence of a static magnetic field. It is utilized in astrophysics to measure the magnetic fields of stars and sunspots, not the kinematic movement of galaxies.
- **Photoelectric effect:** The photoelectric effect refers to the emission of electrons from a material when illuminated by electromagnetic radiation. It provides foundational evidence for the quantum (particle) nature of light but offers no observational data regarding the macroscopic dynamics of the universe.
- **Additional Facts:**
 - ✓ **Cosmological Redshift vs. Kinematic Doppler Shift:** While historically introduced as a Doppler effect, the redshift observed in distant galaxies is more precisely termed a cosmological redshift. It is caused not by galaxies moving through pre-existing space, but by the continuous metric expansion of spacetime itself, which stretches the wavelength of traveling photons.
 - ✓ **Hubble's Law:** The observation of this redshift led to the formulation of Hubble's Law. This law states that the recessional velocity of a galaxy is directly proportional to its proper distance from the observer (expressed simply as velocity equals the Hubble constant multiplied by distance).
 - ✓ **Blueshift:** Conversely, if an astronomical object is moving toward the observer due to local gravitational attraction, the Doppler effect compresses the observed wavelengths, resulting in a blueshift. For example, the Andromeda Galaxy exhibits a blueshift because it is on a collision course with the Milky Way.

- ✓ **Cosmic Microwave Background:** Alongside galactic redshift, the Cosmic Microwave Background (CMB) constitutes the other primary pillar of observational evidence for an expanding universe. The CMB represents the relic thermal radiation from the early universe, which has been stretched into the microwave region of the spectrum by billions of years of cosmic expansion.

Q 58. Ans. B

Explanation:

- Multi-Agent Systems operate by employing a primary coordinating agent (often called an orchestrator or router) that breaks down complex, multi-step workflows into discrete sub-tasks. These sub-tasks are then delegated to specialized autonomous agents, each optimized for a specific function (such as code generation, data retrieval, or mathematical verification), which significantly improves the accuracy and efficiency of the overall output.

Q 59. Ans. D

Explanation:

- The primary vision for IMT-2030 (6G) encompasses the deep integration of traditional communication networks with advanced environmental sensing, pervasive artificial intelligence computing, and non-terrestrial networks, creating a ubiquitous and intelligent global ecosystem.
- **Additional Facts:**
 - ✓ The International Telecommunication Union officially designated the framework for 6G as “IMT-2030,” indicating that the foundational standards and preliminary commercial deployments are widely anticipated to materialise around the year 2030.
 - ✓ Unlike previous network generations where artificial intelligence was applied retroactively as an optimisation tool, 6G architectures are being designed with machine learning natively embedded at the physical layer to autonomously manage network traffic, power consumption, and spectral efficiency.

Q 60. Ans. C

Explanation:

- Solid-State Battery (SSB) technology is currently advancing through pilot-scale manufacturing and vehicle validation phases. Material scientists are

actively engineering solutions for primary technical barriers, specifically the high electrical resistance at the solid-to-solid interfaces and the propagation of microscopic lithium dendrites that can pierce the solid electrolyte and cause short circuits.

Q 61. Ans. C

Explanation:

- State-of-the-art multimodal models utilise joint representation spaces and cross-modal attention mechanisms. This framework allows the core transformer architecture to simultaneously process, align, and correlate tokens originating from diverse modalities, enabling a unified and deeply contextual understanding of the input.

Q 62. Ans. A

Explanation:

- Hybrid quantum-classical computing involves an architecture where quantum processors work in tandem with classical computational systems. The classical systems handle control orchestration, data pre-processing, algorithmic optimization routines, and error correction, while delegating specific, highly complex mathematical calculations to the quantum processor.

Q 63. Ans. B

Explanation:

- Pataneshwar Dham is a historical Hindu temple situated on the banks of the Kiul River in the Jamui district of Bihar.
- Pataneshwar Dham is primarily dedicated to Lord Shiva.
- The temple complex is geographically positioned on a hill directly adjacent to the Kiul River, a tributary of the Ganges, which provides a natural source of water for temple rituals.

Q 64. Ans. B

Explanation:

- Professor R. K. Sinha, an academic and environmentalist from Bihar, is globally recognized as the “Dolphin Man of India.” He earned this title for his decades of pioneering research and dedicated conservation efforts to protect the endangered Ganges river dolphin (*Platanista gangetica*).
- **Additional Facts:**
 - ✓ **Waterman of India:** Rajendra Singh
 - ✓ **Birdman of India:** Salim Ali
 - ✓ **Forest Man of India:** Jadav Payeng

- ✓ **Tiger Man of India:** Kailash Sankhala
- ✓ **Mountain Man of India:** Dashrath Manjhi (also from Bihar)
- ✓ **Metro Man of India:** E. Sreedharan
- ✓ **Missile Man of India:** Dr. A.P.J. Abdul Kalam
- ✓ **Missile Woman of India:** Tessy Thomas
- ✓ **Cyclone Man of India:** Mrutyunjay Mohapatra

Q 65. Ans. B

Explanation:

- **Statement 1 is correct:** Bihar operates under a three-tier Panchayati Raj system in accordance with the Bihar Panchayat Raj Act, 2006. This structure consists of the Gram Panchayat at the village level, the Panchayat Samiti at the block or intermediate level, and the Zila Parishad at the district level.
- **Statement 2 is correct:** As mandated by Article 243K of the Indian Constitution, the Bihar State Election Commission is an independent constitutional body strictly responsible for the superintendence, direction, and control of all elections to the Panchayati Raj Institutions and urban local bodies within the state. (The Election Commission of India conducts state assembly and national elections, not local panchayat elections).
- **Statement 3 is incorrect:** Bihar was not the first state to implement the 73rd Constitutional Amendment. Madhya Pradesh holds the distinction of being the first state to conduct elections under the new provisions of the 73rd Amendment in 1994. Historically, Rajasthan was the first state to establish the Panchayati Raj system (in Nagaur district, 1959), well before the constitutional amendment was enacted.
- **Additional Facts:**
 - ✓ **Women's Reservation:** While it was not the first to implement the 73rd Amendment, Bihar did make history in 2006 by becoming the first state in India to implement a strict 50 percent reservation for women in all Panchayati Raj Institutions.
 - ✓ **EBC Reservation:** The Bihar Panchayat Raj Act of 2006 also mandated a 20 percent reservation for Extremely Backward Classes (EBCs) across all three tiers of the local self-government system.
 - ✓ **Gram Katchahry:** Bihar's system includes a unique judicial mechanism at the village level known as the Gram Katchahry. Headed by an elected Sarpanch, it operates alongside the Gram Panchayat to mediate and resolve minor civil and criminal disputes locally.

- ✓ **Tenure and Dissolution:** The constitutional tenure of every Panchayat is exactly five years from the date of its first meeting. If a Panchayat is dissolved prematurely, mandatory elections must be conducted within six months of its dissolution to elect a new body for the remainder of the term.

Q 66. Ans. B

Explanation:

- The Kesariya Stupa is located in the East Champaran district of Bihar. Standing at a height of approximately 104 feet with a circumference of nearly 400 feet, it is considered the tallest and largest Buddhist stupa in the world.
- **Additional Facts:**
 - ✓ **Vishwa Shanti Stupa (Peace Pagoda):** Rajgir (Nalanda district).
 - ✓ **Buddha Relic Stupa:** Vaishali (Vaishali district).
 - ✓ **Sujata Stupa:** Bodh Gaya (Gaya district).
 - ✓ **Mahabodhi Temple Complex:** Bodh Gaya (Gaya district).
 - ✓ **Lauriya Nandangarh Stupa:** West Champaran district.

Q 67. Ans. A

Explanation:

- Purushottam Das Tandon was elected as the President of the Bihar Provincial Kisan Sabha in 1934. Specifically, he presided over the Second Bihar Provincial Farmers Conference, which was held in Gaya on August 29-30, 1934.
- **Additional Facts:**
 - ✓ **Second Provincial Conference:** The 1934 conference in Gaya, presided over by Tandon, was a major event intended to mobilize peasants against colonial exploitation, excessive land revenues, and the oppressive practices of local zamindars.
 - ✓ **Prominent Attendees:** The conference served as a platform for national unity, drawing the attendance of several prominent freedom fighters from outside Bihar, notably Khan Abdul Ghaffar Khan (popularly known as the "Frontier Gandhi").
 - ✓ **Role of Swami Sahajanand:** Swami Sahajanand Saraswati was the foundational leader of the peasant movement in Bihar and the original founder of the Bihar Provincial Kisan Sabha in 1929. However, he invited Tandon to preside over

the 1934 session to help bridge regional agrarian movements and build national solidarity.

- ✓ Although primarily a political figure from Uttar Pradesh (where he founded the Allahabad District Peasants' Committee in 1918), Purushottam Das Tandon was widely respected across state for his relentless advocacy for farmers, making him a natural choice for the Bihar presidency.
- ✓ Tandon was deeply revered and customarily given the title "Rajarshi" (Royal Saint). A fierce advocate for Hindi as India's official language, he served in the Constituent Assembly and was awarded the Bharat Ratna in 1961.

Q 68. Ans. B

Explanation:

- The Dhruva Dal was formed during the Quit India Movement in 1942. On August 13, 1942, a 13-year-old boy named Dhruv Kundu bravely hoisted the Indian national flag at the Katihar police station in Bihar. He was fired upon by British soldiers and succumbed to his injuries a couple of days later. In retaliation and to honor his sacrifice, local revolutionaries formed a resistance group named the "Dhruva Dal".
- **Additional Facts:**
 - ✓ Prominent freedom fighter Jayaprakash Narayan (JP) was closely associated with this underground revolutionary group and was made the supreme commander of the "Dhruva Dal" to guide the militant youth resistance in the region.
 - ✓ The Quit India Movement, launched by Mahatma Gandhi on August 8, 1942, is also known as the August Revolution. After major national leaders were immediately arrested, it became a largely leaderless and highly aggressive movement, particularly in regions like Bihar and the United Provinces.
 - ✓ **Shahid Chowk:** To honor the brave activists who lost their lives in Katihar during the 1942 movement, including young Dhruv Kundu, the city's central square was subsequently named "Shahid Chowk" (Martyrs' Square).

Q 69. Ans. C

Explanation:

- **India Divided** was not authored by Jayaprakash Narayan. It was written by Dr. Rajendra Prasad, who later became the first President of independent India. Published in 1946, it is a detailed, critical

analysis of the two-nation theory and the impending partition of India.

● Additional Facts:

- ✓ **Jayaprakash Narayan (Loknayak):** Often referred to as "The People's Hero," Jayaprakash Narayan was a prominent Indian independence activist, socialist theorist, and political leader. He is best remembered for leading the mid-1970s opposition against Prime Minister Indira Gandhi, for whose overthrow he called for a "Total Revolution."
- ✓ **Sampoorna Kranti Movement:** The Total Revolution was not merely political; JP envisioned a seven-pronged transformation encompassing social, economic, political, cultural, ideological, educational, and moral changes in Indian society.
- ✓ **Other Major Works by JP:** Some of Jayaprakash Narayan's other highly influential writings include *Why Socialism?* (1936), *Towards Struggle* (1946), and his famous *Prison Diary* written during the Emergency in 1975.
- ✓ **Other Works by Dr. Rajendra Prasad:** Besides *India Divided*, Dr. Rajendra Prasad was a prolific writer. His other notable books include *Satyagraha in Champaran*, *Atmakatha* (his autobiography), and *Mahatma Gandhi and Bihar, Some Reminiscences*.

Q 70. Ans. B

Explanation:

- The **Portuguese** were the **first** European power to establish a commercial presence in Bihar. Entering the region in the early 17th century, their merchants navigated up the Ganges River from their base in Hooghly, Bengal, to trade in Patna. They brought spices and porcelain to exchange for Bihar's cotton textiles and saltpetre.
- The **British** were the second European power to enter Bihar. They attempted to set up their first factory for saltpetre trade at Alamganj in Patna in 1620, but it was closed the following year and only successfully revived in 1651.
- The **Dutch** East India Company arrived after the Portuguese and the English. They established their first permanent factory in Patna in 1632 (the building currently houses Patna College), focusing heavily on the saltpetre, grain, and cotton trade.
- The **French** arrived much later, establishing their presence in the region toward the late 17th century to compete with the British and Dutch.

Q 71. Ans. D

Explanation:

- **Swami Yogananda** was not associated with the formation of the Bihar Socialist Party. Instead, he was a prominent peasant leader associated with the Kisan Sabha movement in Bihar, working alongside Swami Sahajanand Saraswati to organize farmers.
- **Rambriksh Benipuri**: He was one of the key founding members of the Bihar Socialist Party in 1931, playing a significant role in organizing the party and propagating socialist ideas.
- **Phulan Prasad Verma**: He was a prominent founder of the Bihar Socialist Party. Alongside Jayaprakash Narayan and others, he helped establish its ideological foundation.
- **Ganga Sharan Sinha**: He was also a founding member of the Bihar Socialist Party in 1931 and served as one of its first secretaries.
- **Additional Facts:**
 - ✓ **Foundation**: The Bihar Socialist Party was formed in July 1931 by Marxist-Leninist socialists within the Indian National Congress. Its primary organizers included Ganga Sharan Sinha, Rambriksh Benipuri, Phulan Prasad Verma, and Ramanand Mishra.
 - ✓ **Initial Leadership**: When the organizing committee was established, Abdul Bari served as the President, Ganga Sharan Sinha and Phulan Prasad Verma as Secretaries, and Ambika Kant Sinha as the Treasurer.
 - ✓ **Objectives**: The party's core objectives were the propagation of socialist ideas and the formation of robust peasant and labor organizations.
 - ✓ **Precursor to CSP**: The ideological groundwork and organizational structure of the Bihar Socialist Party later provided the foundation for the broader Congress Socialist Party (CSP), which was officially formed in 1934.

Q 72. Ans. A

Explanation:

- **Rohtas**: The Amjhore Hills, renowned for their extensive pyrite deposits, are situated in the Rohtas district of Bihar. This region holds the distinction of having some of the largest pyrite reserves in India.
- **Kaimur**: While Kaimur is rich in mineral resources, particularly limestone which supports the local cement industry, it is not the location of the Amjhore pyrite deposits.

- **Nawada**: Nawada district is historically associated with mica deposits, largely due to its geographical proximity to the Koderma mica belt, rather than pyrites.

- **Aurangabad**: Aurangabad is known for its historical and religious sites, such as the Deo Sun Temple, and some coal occurrences, but it does not host the Amjhore pyrite mines.

Additional Facts:

- ✓ **Fool's Gold**: Pyrite is an iron sulfide mineral. Due to its metallic luster and pale brass-yellow hue, it is popularly referred to as "fool's gold."
- ✓ **National Monopoly**: Bihar holds a near-monopoly on pyrite reserves in India, with the vast majority (over 90%) concentrated in the Amjhore region of the Rohtas district.
- ✓ **Industrial Application**: The primary industrial use of the pyrite mined at Amjhore is the production of sulfuric acid, which is a critical component in the manufacturing of agricultural fertilizers.
- ✓ **PPCL**: The Government of India established Pyrites, Phosphates and Chemicals Limited (PPCL) at Amjhore to systematically mine the ore and produce fertilizers. Although the factory is currently defunct, it played a major role in the regional economy for several decades.

Q 73. Ans. C

Explanation:

- The Jal Jeevan Hariyali Mission was officially launched by Bihar Chief Minister Nitish Kumar on **August 9, 2019**, directly coinciding with the occasion of 'Bihar Prithvi Diwas' (Bihar Earth Day). Following this launch, the state cabinet formally approved a budget of Rs 24,524 crore to implement the campaign in a rigorous mission mode over three financial years.

Additional Facts:

- ✓ **Implementation Date**: While the official launch announcement was made on August 9, 2019, the mission was actively implemented across the state starting October 2, 2019, to commemorate the 150th birth anniversary of Mahatma Gandhi.
- ✓ **Core Philosophy**: The underlying motto of the mission is "Water, Life and Greenery — only then will be Prosperity," emphasising that human life and prosperity are impossible without water and ecological balance.

- ✓ **11-Point Agenda:** The mission is fundamentally driven by an 11-point agenda that focuses on restoring public water structures (ponds, wells, and check dams), constructing rainwater harvesting systems in public buildings, transferring water to deficit areas (such as the Ganga Water Supply Scheme to Gaya and Rajgir), and promoting solar energy.
- ✓ **Green Cover Increase:** A major success of the mission is its afforestation drive. After Bihar's bifurcation in 2000, its green cover had fallen to just 9%. Through initiatives like Jal Jeevan Hariyali, the state's green cover has been successfully increased to over 15%, with continued targets to exceed 17%.

Q 74. Ans. C

Explanation:

- **Statement 1 is correct:** The Someshwar Pass is a natural gorge formed by the continuous erosional action of the Turipani (or Juripani) river flowing down from Nepal.
- **Statement 2 is correct:** The Bhikhna Thori Pass is created by the river action of the Kudi river.
- **Statement 3 is correct:** The Marwat Pass is formed by the flow and erosion of the Harha river.
- All three of these major natural passes cut through the Someshwar Range and have historically functioned as critical trade and communication routes connecting the Indian state of Bihar with neighboring Nepal.
- **Additional Facts:**
 - ✓ **Geographical Location:** The Someshwar Range is situated in the West Champaran district. It is an extension of the Shiwalik Range, which constitutes the outermost and youngest foothills of the Himalayas.
 - ✓ **Highest Point in Bihar:** The Someshwar Range contains the Someshwar Fort. At an altitude of 874 meters, it is the highest topographical point in the entire state of Bihar.
 - ✓ **Vulnerability to Erosion:** The hills in this region are primarily composed of soft, sedimentary rocks from the Tertiary period, such as sandstone, clay, and conglomerates. Because these materials are relatively soft, they are highly susceptible to being carved out by fast-flowing rivers, resulting in the deep passes.
 - ✓ **Regional Sub-divisions:** Geographically, the Shiwalik region in Bihar is divided into three distinct sub-regions: the Someshwar Range

(the northernmost boundary), the Ramnagar Dun (a broken hilly tract to the south), and the Harha Valley (also known as Dun Valley), which is nestled between the two.

Q 75. Ans. C

Explanation:

- **Statement A is correct:** The Kosi division consists of exactly three districts: Saharsa (its headquarters), Supaul, and Madhepura.
- **Statement B is correct:** The Purnia division is located in the northeastern part of the state (the Seemanchal region) and comprises four districts: Purnia, Katihar, Araria, and Kishanganj.
- **Statement C is incorrect:** The Saran division includes only three districts: Saran (Chhapra), Siwan, and Gopalganj. Madhubani is not a part of the Saran division; it falls under the jurisdiction of the Darbhanga division (along with Darbhanga and Samastipur).
- **Statement D is correct:** The Tirhut division, with its headquarters in Muzaffarpur, is one of the largest administrative divisions in Bihar and comprises six districts: East Champaran, West Champaran, Muzaffarpur, Sitamarhi, Sheohar, and Vaishali.
- **Additional Facts**
 - ✓ **Administrative Setup:** Bihar is administratively divided into 9 divisions, which are further subdivided into a total of 38 districts, 101 subdivisions, and 534 community development blocks.
 - ✓ **The 9 Divisions:** The nine divisions of Bihar are Patna, Tirhut, Saran, Darbhanga, Kosi, Purnia, Bhagalpur, Munger, and Magadh.
 - ✓ **Largest and Smallest Divisions:** The Tirhut, Munger, and Patna divisions are the largest, each comprising six districts. Conversely, the Bhagalpur division is the smallest, consisting of only two districts: Bhagalpur and Banka.

Q 76. Ans. B

Explanation:

- Arwal was officially created on **August 20, 2001**, by bifurcating the **Jehanabad district**. With its formation, it became the 38th and newest district in the state of Bihar.
- **Additional Facts:**
 - ✓ **Administrative Division:** Arwal falls under the jurisdiction of the Magadh administrative division, which is headquartered in Gaya. The other districts in this division are Gaya, Nawada, Aurangabad, and Jehanabad.

- ✓ **Creation Rationale:** The district was primarily established to improve administrative efficiency, enhance local governance, and better manage law and order. Historically, this specific region was severely affected by left-wing extremism (Naxalism), prompting the need for a more localized administrative center.
- ✓ **Geographical Boundaries:** The district is naturally bounded on its western side by the Sone River. This river acts as a vital physical border separating Arwal from the neighboring Bhojpur and Rohtas districts.
- ✓ **Economy and Agriculture:** The economy of Arwal is almost entirely agrarian. Because the district lacks major industrial infrastructure, the local population relies heavily on agriculture, which is largely supported by the intricate canal irrigation systems fed by the Sone River.

Q 77. Ans. C

Explanation:

- Araria does not share a border with West Bengal. It is geographically blocked from the West Bengal border by the districts of Kishanganj (to its east) and Purnia (to its south). Instead, Araria shares an international border with Nepal to its north.
- **Additional Facts:**
 - ✓ **West Bengal Border:** Only three districts of Bihar share a border with West Bengal: Kishanganj, Purnia, and Katihar.
 - ✓ **International Border:** Bihar shares its northern international border exclusively with Nepal. Seven districts touch this border: West Champaran, East Champaran, Sitamarhi, Madhubani, Supaul, Araria, and Kishanganj.
 - ✓ **Uttar Pradesh Border:** To the west, Bihar shares a border with Uttar Pradesh. Eight districts touch UP: West Champaran, Gopalganj, Siwan, Saran, Bhojpur, Buxar, Kaimur, and Rohtas.
 - ✓ **Jharkhand Border:** To the south, Bihar shares a border with Jharkhand (which was carved out of Bihar in 2000). Eight districts share this boundary: Rohtas, Aurangabad, Gaya, Nawada, Jamui, Banka, Bhagalpur, and Katihar.

Q 78. Ans. B

Explanation:

- Someshwar Hill, along with the entire Someshwar Range, is located in the northernmost part of the **West Champaran** district. This range forms the international boundary between Bihar (India) and Nepal.

● Additional Facts:

- ✓ **Geological Origin:** The Someshwar Range is a part of the Shiwalik hills, representing the outermost and youngest foothills of the Himalayas. They are primarily composed of tertiary sedimentary rocks like sandstone and conglomerates.
- ✓ **Highest Altitude:** The peak of the Someshwar Hill, where the ruins of the Someshwar Fort are located, stands at approximately 874 meters (2,867 feet) above sea level, making it the highest geographical point in Bihar.
- ✓ **Ecological Significance:** The foothills of the Someshwar Range are covered by dense deciduous forests and constitute a major part of the Valmiki Tiger Reserve and National Park.
- ✓ **Drainage:** The slopes of this range are drained by several fast-flowing, erosion-heavy rivers and streams, which have carved out the natural passes (like Someshwar, Bhikhna Thori, and Marwat) that facilitate movement between Bihar and Nepal.

Q 79. Ans. D

Explanation:

- The Shiwalik Hills in Bihar (and across the entire Himalayan foothills) are predominantly composed of tertiary sedimentary rocks. They were formed over millions of years by the accumulation and consolidation of sand, clay, pebbles, and conglomerates brought down by fast-flowing Himalayan rivers.
- **Additional Facts:**
 - ✓ **Geological Age:** The Shiwalik rocks belong to the Tertiary period. They are the youngest geological formations in Bihar, contrasting sharply with the ancient Precambrian rocks found in the southern parts of the state.
 - ✓ **Physical Characteristics:** Because they are sedimentary and relatively young, these rocks are generally soft, unconsolidated, and highly susceptible to weathering and rapid river erosion. This softness is what allows the local rivers to carve the deep passes (like Bhikhna Thori) through the hills.
 - ✓ **Location in Bihar:** The Shiwalik rock system in Bihar is exclusively confined to the northwestern corner of the state, specifically the northern part of the West Champaran district.

✓ **Other Rock Systems in Bihar:**

- **Vindhyan System:** Found in southwestern Bihar (Rohtas and Kaimur), primarily consisting of limestone, sandstone, and shale (also sedimentary, but much older and harder).
- **Dharwar System:** Found in southeastern Bihar (Munger, Jamui, Nawada), known for ancient metamorphic and igneous rocks containing minerals like mica.
- **Quaternary System:** The vast Gangetic alluvial plains that make up the majority of Bihar's landmass.

Q 80. Ans. A

Explanation:

- The Udaipur Wildlife Sanctuary is located in the West Champaran district of Bihar. While the district of West Champaran is globally renowned for its tiger habitat (specifically the Valmiki Tiger Reserve), the Udaipur Wildlife Sanctuary itself is primarily an oxbow lake wetland famous for its avian population.
- **Additional Facts**
 - ✓ **Ecological Profile:** The Udaipur Wildlife Sanctuary is predominantly a wetland situated in an oxbow lake (Sareya Chaur) formed by the Gandak River. It spans an area of approximately 8.74 square kilometers.
 - ✓ **Fauna:** Rather than tigers, the sanctuary is highly celebrated for its rich birdlife. It acts as a crucial wintering ground for numerous migratory birds and is home to local species such as the spot-billed duck, lesser whistling duck, and various waterbirds.
 - ✓ **Valmiki Tiger Reserve:** The tiger habitat referenced in the region belongs to the Valmiki National Park and Tiger Reserve, which is also located in the West Champaran district and forms the core of Bihar's tiger conservation efforts.
 - ✓ **Other Notable Sanctuaries in Bihar:**
 - **Gautam Buddha Wildlife Sanctuary:** Located in Gaya.
 - **Vikramshila Gangetic Dolphin Sanctuary:** Located in Bhagalpur; India's only sanctuary dedicated to the endangered Gangetic dolphin.
 - **Nagi Dam and Nakti Dam Bird Sanctuaries:** Located in Jamui.

Q 81. Ans. B

Explanation:

- Spread across the Kaimur and Rohtas districts, the **Kaimur Wildlife Sanctuary** is the largest in Bihar, covering an extensive geographical area of approximately 1,342 square kilometers.
- **Additional Facts:**
 - ✓ **Geographical Features:** The Kaimur Wildlife Sanctuary is situated in the Kaimur Range, which is an eastern extension of the Vindhya Range. The landscape is characterized by undulating hills, dense deciduous forests, and stunning waterfalls like Karkat Waterfall and Telhar Kund.
 - ✓ **Second Tiger Reserve:** The National Tiger Conservation Authority (NTCA) has given in-principle approval to declare the Kaimur Wildlife Sanctuary as Bihar's second Tiger Reserve (after Valmiki).
 - ✓ **Fauna:** The sanctuary acts as a critical corridor for wildlife movement between the forests of Bihar, Jharkhand, Uttar Pradesh, and Madhya Pradesh. It is home to a significant population of leopards, sloth bears, chinkaras, nilgais, and various species of deer.
 - ✓ **Historical Sites:** The sanctuary region also encompasses the ancient Rohtasgarh Fort and Shergarh Fort, blending rich historical heritage with ecological conservation.

Q 82. Ans. B

Explanation:

- Prime Minister **Narendra Modi** officially launched the Bihar Rajya Jeevika Nidhi Saakh Sahkari Sangh Limited via video conferencing on September 2, 2025. The initiative is designed to provide rural women entrepreneurs associated with Jeevika in Bihar with easy access to affordable credit, operating entirely on a digital platform to ensure transparent fund transfers.

Q 83. Ans. D

Explanation:

- In June 2025, the Bihar Government announced a major policy revision, increasing the monthly social security pension from the previous ₹400 to **₹1,100**. Effective from July 2025, this hike benefits over 1.09 crore people across the state, including senior citizens (under schemes like the Mukhyamantri Vridhjan Pension Yojana), persons with disabilities, and widowed women (under the Lakshmi Bai Pension Yojana).

● **Additional Facts:**

- ✓ **Payment Mechanism:** To ensure transparency and timeliness, the enhanced pension amount is directly transferred to the beneficiaries' bank accounts on the 10th of every month via the Direct Benefit Transfer (DBT) system.
- ✓ **Automatic Upgrade:** Existing beneficiaries registered on the state portal did not need to reapply to receive the new amount, as the ₹700 revision was applied automatically to all active accounts.
- ✓ **Electoral Context:** This substantial monthly increase was a key welfare measure announced by Chief Minister Nitish Kumar just months ahead of the 2025 Bihar Assembly elections.

Q 84. Ans. A

Explanation:

- The Government of Bihar has launched a major grassroots governance initiative, deciding to observe 'Panchayat Vikas Diwas' (Village Development Day) across all 8,500+ gram panchayats in the state on the **last Sunday of every month**.
- The primary objective of this initiative is to realise Mahatma Gandhi's vision of 'Gram Swaraj' by strengthening local governance, accelerating rural development, and promoting transparency and public participation.
- During these monthly sessions, elected representatives (including MPs, MLAs, and Panchayat members) and government officials hold Gram Sabhas to deliberate on local issues, track the progress of development schemes on portals like e-Gram Swaraj, publicly review panchayat funds, and collectively listen to the Prime Minister's 'Mann Ki Baat' radio broadcast.
- Each month's Panchayat Vikas Diwas features a specific theme to guide discussions, with the inaugural edition focusing on the theme of a 'Women-Friendly Panchayat' (Mahila Hitaishi Panchayat).

Q 85. Ans. C

Explanation:

- In February 2026, Bihar was officially declared completely "Naxal-free" following the surrender of Suresh Koda, a top Maoist commander, before the Special Task Force (STF) in the Munger district. This milestone marked the end of active armed Maoist squads operating within the state.
- **Additional Facts:**
 - ✓ **Suresh Koda:** Also known as Mustakim,

Koda was a self-styled commander of the Bihar-Jharkhand Special Zonal Committee of the banned CPI (Maoist). He had been evading arrest for 25 years, was wanted in approximately 60 cases (including UAPA charges), and carried a bounty of ₹3 lakh.

- ✓ Under the Bihar government's 'Atmasmarpan Sah Punarvaas Yojana' (Surrender-cum-Rehabilitation Scheme), Koda is entitled to receive his own ₹3 lakh bounty amount, an additional incentive of ₹5 lakh, and a monthly stipend of ₹10,000 for three years to pursue vocational training.

Q 86. Ans. C

Explanation:

- In December 2025, the Bihar State Power Generation Company Ltd (BSPGCL) signed MoUs worth ₹13,000 crore for two massive pumped storage power projects in the **Nawada district**. The projects, which boast a combined capacity of 2,120 MW, are being developed by GreenCo Energies and Sun Petrochemicals in the hilly terrains of Ekamba and Gosaitari villages.

Q 87. Ans. C

Explanation:

- The newly formed Bihar cabinet approved a comprehensive action plan to establish **25 new sugar mills**, alongside reviving **9 currently closed sugar mills** across the state. This aggressive push is a core component of the government's "Saat Nischay-3" (Seven Resolves) roadmap and the "New Age Economy" framework. The move is designed to attract substantial investments, boost rural employment, and restore Bihar as a leading sugarcane-producing belt.

Q 88. Ans. A

Explanation:

- According to the report of the 16th Finance Commission (covering the award period 2026-31), Bihar's designated share in the divisible pool of central taxes is **9.95%**. This represents a slight decline from the 10.06% share recommended by the previous 15th Finance Commission.

Q 89. Ans. B

Explanation:

- Under the Mukhyamantri Bihar Heli-Tourism and Air Tourism Service Scheme 2026, the state

government has set a highly subsidized fare of ₹2,100 per passenger for a 10-minute helicopter joyride over Patna. Operating on weekends, the flights give tourists an aerial view of major landmarks like the Ganga River, JP Ganga Path, and Gandhi Maidan.

Q 90. Ans. A

Explanation:

- The National Data Centre for the North Eastern Region (NDC-NER) was inaugurated by Prime Minister Narendra Modi at Amingaon in the **Kamrup district of Assam**. This state-of-the-art facility acts as the core digital infrastructure backbone for all eight Northeastern states.
- **Additional Facts:**
 - ✓ **Infrastructure Details:** Built by Yotta Data Services on behalf of the National Informatics Centre (NIC), the facility is a Tier III, IGBC Gold-rated greenfield project. It is spread across a 43,000 sq ft campus and is specially engineered with deep load-bearing piles to withstand Seismic Zone V conditions and high groundwater levels.
 - ✓ **Power and Scale:** The facility currently houses 200 racks with a 2 MW IT load, designed to scale up to an 8 MW total power capacity and 400 high-density racks to meet future demands for cloud and AI workloads.
 - ✓ **Digital Sovereignty:** A primary goal of the NDC-NER is to ensure that sensitive government data, citizen-facing services, and e-governance platforms remain secure, reliable, and entirely localized within India's borders.

Q 91. Ans. B

Explanation:

- **Pacific Ocean (Correct):** The Coral Sea is a marginal sea of the South Pacific Ocean, located off the northeast coast of Australia. It is bounded by the eastern coast of Australia, Vanuatu, the Solomon Islands, and the southern coast of New Guinea. It is internationally renowned for housing the world's most extensive coral reef ecosystem.
- **Additional Facts:**
 - ✓ **The Great Barrier Reef:** The Coral Sea is home to the Great Barrier Reef, the world's largest coral reef system. It stretches over 2,300 kilometers and can be seen from space, comprising roughly 2,900 individual reefs and 900 islands.

- ✓ **The Battle of the Coral Sea (1942):** During World War II, this region was the site of a major strategic naval engagement between the Imperial Japanese Navy and the Allied forces (United States and Australia). It is historically significant as the first naval battle in history fought entirely by aircraft carriers, where the ships never saw or fired directly upon one another.

- ✓ **Coral Sea Marine Park:** Australia manages a vast portion of this area as the Coral Sea Marine Park, which covers nearly 989,842 square kilometers, making it one of the largest marine protected areas in the world.

Q 92. Ans. B

Explanation:

- **Orographic Uplift and Funnelling Effect:** Mawsynram and neighbouring Cherrapunji (Sohra) record the highest average annual rainfall in the world primarily due to orographic rainfall. The South-West monsoon winds sweeping over the Bay of Bengal pick up massive amounts of moisture. As these winds move north, they hit the Meghalaya Plateau and are forced into the funnel-shaped valleys of the Khasi and Jaintia Hills. This physical constriction forces the moisture-laden air to rise rapidly, causing adiabatic cooling, heavy condensation, and intense, prolonged precipitation on the windward slopes.
- **Additional Facts:**
 - ✓ **Geographical Setting:** Mawsynram is situated in the East Khasi Hills district of Meghalaya, at an altitude of roughly 1,400 meters above sea level.
 - ✓ **Annual Rainfall Average:** It receives an average annual rainfall of over 11,800 millimeters (about 467 inches), much of which is concentrated during the monsoon months from June to September.
 - ✓ **Rain-Shadow Effect:** Just a short distance away on the leeward side of these hills (such as towards Shillong or the Brahmaputra valley), the rainfall drops drastically due to the rain-shadow effect, illustrating how sharply topography controls precipitation distribution.

Q 93. Ans. A

Explanation:

- **Am (Tropical Monsoon Climate - Correct):** As per Wladimir Köppen's climatic classification scheme, the windward (western) slope of the Western Ghats (encompassing the Konkan and Malabar coastal belts)

falls under the Am (or Amw) category. This region experiences a tropical monsoon climate characterised by high temperatures throughout the year and very heavy rainfall (exceeding 200–300 cm) concentrated during the summer southwest monsoon months, followed by a short dry winter season.

● **Additional Facts:**

- ✓ **Cwg (Incorrect):** This code denotes a monsoon type with dry winters (humid subtropical climate), which is typically found across most parts of the Ganga Plain, eastern Rajasthan, and the central plateaus.
- ✓ **BShw (Incorrect):** This represents a semi-arid steppe climate. In the context of peninsular India, this zone is found immediately east of the Western Ghats in the rain-shadow region (interior plateau), where moisture levels drop sharply.
- ✓ **Dfc (Incorrect):** This stands for a cold, humid winter climate with short summers, which applies to high-altitude and high-latitude northeastern regions like Sikkim, Arunachal Pradesh, and parts of Assam.

Q 94. Ans. A

Explanation:

- Odisha holds an absolute monopoly in the production and reserves of chromite in India, accounting for nearly 93% to 96% of the country's total chromite resources. The state's Jajpur district—specifically the Sukinda Valley—along with Kendujhar and Dhenkanal, forms the primary hub of India's chromite mining industry.
- **Additional Facts:**
 - ✓ **Chemical Composition:** Chromite is a natural spinel group mineral and is the single commercially viable ore of chromium, chemically expressed as an iron chromium oxide (FeCr_2O_4).
 - ✓ **Key Industrial Uses:** The vast majority of chromite is consumed by the ferro-alloys industry to produce ferrochrome or charge-chrome, which are indispensable components for manufacturing stainless steel, high-temperature engineering alloys, and armour-plating.

Q 95. Ans. A

Explanation:

- **Assertion (A) is true:** *Glossopteris* (a genus of extinct seed ferns) and *Mesosaurus* (a small,

freshwater reptile) are classic examples of fossil evidence. The occurrence of identical terrestrial and freshwater species across continents currently separated by thousands of miles of open ocean was a major geological puzzle before the acceptance of plate tectonics.

● **Reason (R) is true and is the correct explanation:**

The presence of identical fossils on widely separated landmasses can only be logically explained by the fact that these continents were once united as a single contiguous southern supercontinent—Gondwanaland—during the Paleozoic and Mesozoic eras. When Gondwanaland fragmented, the fossil-bearing strata were carried away on separate tectonic plates to their current geographic positions.

● **Additional Facts:**

- ✓ **Pillars of Continental Drift:** Alfred Wegener used fossil distribution as one of the core biological proofs for his Continental Drift Theory (1912), alongside the geometric “jigsaw fit” of continental margins (such as South America and Africa).
- ✓ **Glaciological Evidence:** Another striking proof matching the Gondwanaland hypothesis is the presence of tillite—sedimentary rock formed out of deposits of glaciers—found across southern South America, Africa, Madagascar, India, Antarctica, and Australia, indicating a shared glacial epoch when these lands were clustered near the South Pole.

Q 96. Ans. D

Explanation:

- The Central Place Theory was propounded by the German geographer Walter Christaller in his landmark 1933 publication *Central Places in Southern Germany*. The theory attempts to explain the spatial distribution, size, spacing, and number of cities and towns (settlements) based on the concept of “central places” that provide goods and services to surrounding tributary areas.

Q 97. Ans. C

Explanation:

- **Son River (Correct):** The Son River is one of the largest southern (right-bank) tributaries of the Ganga. It originates on the Amarkantak Plateau in Madhya Pradesh, very close to the source of the Narmada River. It flows northward through a series of gorges and escarpments, crosses into Uttar

Pradesh and Jharkhand, enters Bihar, and finally merges with the Ganga at Maner, located just west of Patna.

● **Additional Facts:**

- ✓ **Major Tributaries of the Son:** The principal tributaries of the Son River are the Rihand and the North Koel rivers, both of which drain major catchment areas in Chhattisgarh, Madhya Pradesh, and Jharkhand.
- ✓ **Dam Projects:** Several vital multi-purpose valley projects have been constructed on its basin, most notably the Bansagar Dam (Madhya Pradesh) on the Son River and the Rihand Dam on its tributary, which creates the Govind Ballabh Pant Sagar reservoir.
- ✓ **Drainage Pattern:** The Son is a classic example of a peninsular river that flows northward and joins an east-flowing Himalayan river system (the Ganga), contrasting with rivers like the Narmada and Tapi which flow westward into the Arabian Sea from the same general highland core.

Q 98. Ans. B

Explanation:

- **Troposphere (Correct):** The troposphere is the lowest layer of Earth's atmosphere, extending from the surface up to an average height of about 8 kilometres at the poles and 18 kilometres at the equator. It contains roughly 75% to 80% of the atmosphere's mass and almost all the atmospheric water vapor and dust particles. Because of this high concentration of moisture and thermal instability (temperature decreases with altitude at the normal lapse rate), virtually all weather phenomena—including cloud formation, precipitation, condensation, and thunderstorms—take place exclusively in this layer.
- **Additional Facts:**
 - ✓ **Stratosphere:** Located above the troposphere, it contains the ozone layer and is largely free from weather, making it suitable for commercial jet aircraft.
 - ✓ **Mesosphere:** Situated above the stratosphere, the mesosphere is the coldest layer of the atmosphere where incoming meteors typically burn up upon entry.
 - ✓ **Exosphere:** The outermost upper boundary layer of Earth's atmosphere, where the atmosphere gradually thins out and merges into the vacuum of space.

Q 99. Ans. B

Explanation:

- **Geostrophic Wind (Correct):** In the upper troposphere, where the friction from Earth's surface is negligible, the movement of air is primarily controlled by two forces: the Pressure Gradient Force (PGF), which pushes air from high pressure to low pressure, and the Coriolis Force, which deflects moving air due to Earth's rotation. When these two opposing forces balance each other out exactly, the resultant wind flows parallel to the straight isobars. This steady-state wind is known as the geostrophic wind.
- **Additional Facts:**
 - ✓ **Direction of Deflection:** Due to the Coriolis effect, the Coriolis force acts perpendicular to the wind direction—to the right in the Northern Hemisphere and to the left in the Southern Hemisphere. This is why the wind does not blow straight across isobars from high to low pressure in the upper atmosphere.
 - ✓ **Buys Ballot's Law:** This law states that in the Northern Hemisphere, if a person stands with their back to the wind, lower pressure is to their left and higher pressure is to their right (and vice versa in the Southern Hemisphere), reflecting the behaviour of geostrophic flow.

Q 100. Ans. D

Explanation:

- **S-waves:** These travel only through solid materials. Since the outer core is liquid, S-waves cannot pass through it and are absent beyond 105° from the epicentre.
- **P-waves:** These can travel through both solids and liquids, but they are strongly refracted at the mantle-outer core boundary, creating a P-wave shadow zone between 105° and 145°.
- **Both Absent:** As S-waves are absent beyond 105° and P-waves are absent only between 105° and 145°, this region forms the shadow zone where both P-waves and S-waves are not recorded. Beyond 145°, P-waves reappear, while S-waves remain absent.

Q 101. Ans. D

Explanation:

- **Statement 1 is correct:** A Parliamentary Committee is one that is appointed or elected by the House or nominated by the Speaker (in the Lok Sabha) or the

Chairman (in the Rajya Sabha). It operates under their direction, and submits its final report either to the House or directly to the Speaker/Chairman.

- **Statement 2 is incorrect:** The recommendations made by Parliamentary Committees are advisory in nature. They are not legally or constitutionally binding on the government. However, the conventions of Parliament require the government to take these recommendations seriously, and ministries are usually required to submit an "Action Taken Report" (ATR) explaining the steps taken or the reasons for not accepting a recommendation.
- **Statement 3 is correct:** Due to the sheer volume of legislative business, technical complexities, and time constraints, the Parliament cannot deeply scrutinize every government action on the floor of the House. Parliamentary Committees act as an extension of Parliament, conducting detailed, rigorous, and largely non-partisan scrutiny of government expenditure, administration, and proposed legislation, thereby enforcing executive accountability to the legislature.

Q 102. Ans. B

Explanation:

- **The Deputy Chairman of the Rajya Sabha (Correct):** According to the Rules of Procedure made by the President under Article 118(3) for joint sittings, the Speaker of the Lok Sabha presides. If the Speaker is absent, the Deputy Speaker of the Lok Sabha presides. If both are absent, the responsibility falls strictly to the Deputy Chairman of the Rajya Sabha.
- **Additional Facts:**
 - ✓ **Triggers for Joint Sitting (Article 108):** A joint sitting is summoned by the President if, after a Bill has been passed by one House and transmitted to the other:
 - The Bill is rejected by the other House; or
 - The Houses have finally disagreed as to the amendments to be made in the Bill; or
 - More than six months elapse from the date of the reception of the Bill by the other House without the Bill being passed by it.
 - **Exceptions:** A joint sitting cannot be summoned for a Money Bill (Article 110) or a Constitutional Amendment Bill (Article 368). It applies only to Ordinary Bills and Financial Bills.
 - ✓ **Quorum and Voting:** The quorum to constitute a joint sitting is one-tenth of the total number of members of both Houses. Decisions are

made by a simple majority of the total number of members of both Houses present and voting. Because of its larger numerical strength, the Lok Sabha generally has an advantage in a joint sitting.

Q 103. Ans. B

Explanation:

- **Article 169 (Correct):** Article 169 of the Constitution of India explicitly provides for the abolition or creation of Legislative Councils (Vidhan Parishad) in States. It vests the ultimate power to create or abolish a State Legislative Council in the Parliament.
- **Simple Majority in Parliament (Correct):** Parliament can enact the law for the creation or abolition of the council by a simple majority (a majority of the members of each House present and voting).
- **Article 170:** Details the composition of the State Legislative Assemblies (Vidhan Sabha), mandating that they consist of not more than 500 and not less than 60 members chosen by direct election.
- **Article 171:** Details the composition of the State Legislative Councils. It states that the total number of members in the council shall not exceed one-third of the total number of members in the state's Legislative Assembly, and shall not be less than 40.
- **Article 172:** Outlines the duration of State Legislatures, specifying a five-year term for the Assembly, while establishing the Council as a permanent body where one-third of its members retire every two years.
- **Additional Facts:**
 - ✓ **Current Bicameral States:** Currently, six Indian states operate with a bicameral legislature (having both an Assembly and a Council): Andhra Pradesh, Bihar, Karnataka, Maharashtra, Telangana, and Uttar Pradesh.
 - ✓ **State Resolution Requirement:** The special majority required in the State Assembly under Article 169 means a majority of the total membership of the Assembly AND a majority of not less than two-thirds of the members present and voting.
 - ✓ **Parliamentary Approval:** Once the state passes the resolution, Parliament must pass a law to create or abolish the council. Importantly, such a law requires only a **simple majority** in Parliament and is not considered a formal constitutional amendment under Article 368.

Q 104. Ans. C

Explanation:

- According to Article 243B(2) of the Constitution of India, introduced by the 73rd Constitutional Amendment Act (1992), a State having a population not exceeding 20 lakhs is exempted from the mandatory requirement of constituting a Panchayat at the intermediate level (often referred to as Panchayat Samiti or Block Panchayat).
- **Two-Tier vs. Three-Tier System:** The 73rd Amendment generally mandates a uniform three-tier Panchayati Raj structure across the country—at the village, intermediate, and district levels. However, to prevent unnecessary administrative and financial burdens on smaller states, this specific exemption allows them to adopt a two-tier system consisting only of the village and district levels.
- **Additional Facts:**
 - ✓ **Part IX and 11th Schedule:** The 73rd Amendment Act added a new Part IX to the Constitution titled “The Panchayats” (spanning Articles 243 to 243O) and a new Eleventh Schedule, which lists 29 functional items placed within the purview of the Panchayats.
 - ✓ **Direct Elections:** All members of Panchayats at the village, intermediate, and district levels must be elected directly by the people (Article 243C). However, the chairpersons at the intermediate and district levels are elected indirectly by and from amongst the elected members thereof.
 - ✓ **Reservation of Seats:** Article 243D mandates the reservation of seats for Scheduled Castes (SCs) and Scheduled Tribes (STs) in proportion to their population. Crucially, not less than one-third of the total number of seats (including those reserved for SC/ST women) must be reserved for women.

Q 105. Ans. A

Explanation:

- **Gram Sabha:** The Provisions of the Panchayats (Extension to the Scheduled Areas) Act, 1996 (PESA) is fundamentally designed to empower the grassroots level. Under PESA, the Gram Sabha is explicitly vested with mandatory executive powers, unlike in non-Scheduled areas where its role is often merely advisory. It is empowered to safeguard community resources, preserve cultural identity, manage minor forest produce, control

local markets, and approve all socio-economic development plans before they are implemented by the Gram Panchayat.

Additional Facts:

- ✓ The PESA Act was enacted based on the recommendations of the Dileep Singh Bhuria Committee (1995). Part IX of the Constitution (73rd Amendment) did not automatically apply to Fifth Schedule areas, so PESA was passed to bridge this gap while respecting tribal self-rule and customs.
- ✓ **Ten States:** Currently, Fifth Schedule areas exist in 10 states: Andhra Pradesh, Telangana, Chhattisgarh, Gujarat, Himachal Pradesh, Jharkhand, Madhya Pradesh, Maharashtra, Odisha, and Rajasthan. All of them have enacted state legislation to give effect to PESA.

Q 106. Ans. B

Explanation:

- **Statement 1 is incorrect:** The President can promulgate an ordinance only when both Houses of Parliament are not in session, or when either of the two Houses is not in session. If both Houses are in session, any law-making must go through the standard legislative route; an ordinance promulgated when both Houses are in session is completely void.
- **Statement 2 is incorrect:** An ordinance cannot be issued to amend the Constitution. The scope of an ordinance is co-extensive with the legislative power of Parliament, meaning it can only cover subjects on which Parliament is competent to make laws (such as those in the Union or Concurrent lists), but it cannot touch the constituent power of amending the Constitution under Article 368.
- **Statement 3 is correct:** The President’s satisfaction regarding the immediate necessity to promulgate an ordinance is not immune from judicial scrutiny. In the landmark *R.C. Cooper vs. Union of India* (1970), the Supreme Court ruled that the President’s satisfaction can be challenged in court on the ground of malafide (bad faith). Although the 38th Constitutional Amendment Act (1975) made the President’s satisfaction final and non-justiciable, the 44th Amendment Act (1978) deleted this clause, fully restoring its judicial review.
- **Statement 4 is correct:** Every ordinance issued by the President must be laid before both Houses of Parliament upon their reassembly. The ordinance ceases to operate at the expiration of six weeks from

the reassembly of Parliament, unless a resolution disapproving it is passed earlier by both Houses, or if it is passed into an Act.

- **Additional Facts:**

- ✓ **Maximum Life of an Ordinance:** Under Article 85, the maximum gap between two sessions of Parliament cannot exceed six months. Because an ordinance can last for six weeks after Parliament reassembles, its absolute maximum survival limit without legislative approval is six months and six weeks.
- ✓ **Discretionary vs. Non-Discretionary:** The President's ordinance-making power is not a discretionary power. The President can only promulgate or withdraw an ordinance on the aid and advice of the Council of Ministers headed by the Prime Minister (Article 74).
- ✓ **Supreme Court on Re-promulgation:** In the D.C. Wadhwa case (1987) and later upheld in the Krishna Kumar Singh case (2017), the Supreme Court held that the repeated re-promulgation of ordinances without submitting them to the legislature is a "subversion of the democratic legislative process" and a "fraud on the Constitution."

Q 107. Ans. A

Explanation:

- **Judicial Review:** This refers to the power of the judiciary (specifically the Supreme Court and High Courts in India) to examine the constitutionality of legislative enactments and executive orders of both the Central and State governments. If any law or executive action is found to violate the provisions of the Constitution, the courts can declare it ultra vires, null, and void, rendering it legally unenforceable.
- **Other Information**
 - ✓ **Judicial Restraint:** This is a philosophy of judicial interpretation where judges are encouraged to limit the exercise of their own power. It asserts that courts should hesitate to strike down laws unless they are patently and absolutely unconstitutional, leaving policy decisions to the democratically elected legislature and executive.
 - ✓ **Judicial Supremacy:** This is a concept where the judiciary has the final and absolute authority to interpret the Constitution and override actions of other branches. Unlike the US system which leans heavily towards judicial supremacy, India features a unique

synthesis of Parliamentary Sovereignty (on the British model) and Judicial Supremacy (on the American model).

- ✓ **Constitutional Amendment:** This is the formal legislative process of modifying or updating the Constitution. Under Article 368, this power is vested strictly in the Parliament, not the judiciary.

- **Additional Facts:**

- ✓ **Implicit vs. Explicit:** The phrase "Judicial Review" is nowhere explicitly mentioned in the Constitution of India. However, the power is clearly implicit in, and derived from, several specific provisions including Articles 13, 32, 131–136, 142, 143, 226, 246, and 372.
- ✓ **Part of the Basic Structure:** In the landmark Kesavananda Bharati case (1973), and subsequently reinforced in the Minerva Mills case (1980), the Supreme Court held that the power of judicial review is a part of the "Basic Structure" of the Constitution. Therefore, it cannot be curtailed, diluted, or excluded even by a constitutional amendment.

Q 108. Ans. B

Explanation:

- **Electoral College of the President (Article 54):** It consists of the elected members of both Houses of Parliament (Lok Sabha and Rajya Sabha), the elected members of the Legislative Assemblies of the States (MLAs), and the elected members of the Legislative Assemblies of the Union Territories of Delhi and Puducherry. Nominated members of Parliament and State Assemblies, as well as all members of State Legislative Councils (MLCs), are strictly excluded.
- **Electoral College of the Vice-President (Article 66):** It consists of both elected and nominated members of both Houses of Parliament. It does not include any members of the State Legislative Assemblies or State Legislative Councils.
- **Comparing the two:**
 - ✓ Nominated members of Parliament participate in the election of the Vice-President but are completely barred from participating in the election of the President.
 - ✓ Elected members of State Legislative Assemblies (and UTs) participate in the President's election but do not participate in the Vice-President's election.
 - ✓ Members of State Legislative Councils (MLCs) do not participate in either election.

Q 109. Ans. D

Explanation:

- **Statement 1 is correct:** The procedure for amending the Constitution under Article 368 can only be initiated by the introduction of a Bill in either House of Parliament (Lok Sabha or Rajya Sabha). State legislatures do not have the power to initiate any Constitutional Amendment Bill.
- **Statement 2 is incorrect:** While it is true that a Constitutional Amendment Bill can be introduced either by a minister or by a private member, it does not require the prior permission or recommendation of the President.
- **Statement 3 is incorrect:** A Constitutional Amendment Bill must be passed by each House of Parliament separately by a special majority. In case of a disagreement between the two Houses, there is no provision for holding a joint sitting (Article 108) to resolve the deadlock, unlike in the case of ordinary bills.
- **Statement 4 is correct:** Once the Bill is duly passed by both Houses (and ratified by the state legislatures, where necessary), it is presented to the President. The President must give his assent to the Bill. He can neither withhold his assent nor return the Bill for reconsideration. This obligation was introduced by the 24th Constitutional Amendment Act of 1971.
- **Additional Facts:**
 - ✓ **Types of Majorities:** Article 368 provides for two types of amendments: (a) By a special majority of Parliament (i.e., a majority of the total membership of each House and a majority of not less than two-thirds of the members of each House present and voting), and (b) By a special majority of Parliament plus ratification by legislatures of not less than one-half of the states by a simple majority.
 - ✓ **Federal Provisions:** If a Constitutional Amendment Bill seeks to amend the federal provisions of the Constitution (such as the election of the President, distribution of legislative powers, or representation of states in Parliament), it must also be ratified by the legislatures of half of the states before being presented to the President.

Q 110. Ans. C

Explanation:

- **Vice President C. P. Radhakrishnan** officially released the updated versions of the Constitution of

India in Tamil and Gujarati on February 21, 2026, to mark International Mother Language Day.

- The release was accompanied by the launch of the 8th Edition of the Legal Glossary in English and Hindi.

Q 111. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The Elephanta Caves are situated on Gharapuri Island in Mumbai Harbour, Maharashtra — famous for their rock-cut Hindu (Shaiva) sculptures, including the iconic Trimurti (Sadashiva) panel, and designated a UNESCO World Heritage Site.
- **Pair 2 is correctly matched:** The Udayagiri Caves are located near Vidisha, Madhya Pradesh (close to Besnagar and Sanchi), and are significant for their Gupta-period rock-cut Hindu sculptures, including the famous Varaha (boar) avatar panel, with inscriptions linking the site to Chandragupta II.
- **Pair 3 is correctly matched:** The Barabar Caves are located in the Makhdumpur region of Jehanabad district, Bihar, approximately 24 km north of Gaya — the oldest surviving rock-cut caves in India, dating to the Mauryan period (322–185 BCE), with several bearing Ashokan inscriptions and famed for their mirror-like “Mauryan polish.”

Q 112. Ans. B

Explanation:

- **Pair 1 is correctly matched:** “Pur charishnu” (also rendered “puracharishnu”) does refer to a “moving fort” in Vedic literature, which is likely an engine or mobile structure used for assaulting strongholds, part of the broader Vedic military vocabulary that also includes “Purandara” (fort-destroyer), an epithet of Indra.
- **Pair 2 is incorrectly matched:** “Goshu” in Vedic society actually refers to the wealth or abundance of cattle, not to “search for cows” or cattle-related conflicts. The term that specifically means “search for cows” (and by extension, war/cattle raids) is “Gavishthi” (sometimes rendered “Goshthi” in its later, non-military sense of “faction/assembly”).
- **Pair 3 is correctly matched:** “Langala” (also spelled langla) does refer to the plough, a key agricultural tool in Vedic society, typically drawn by oxen — with references in the Rigveda to as many as 24 oxen yoked to plough a single field.

Q 113. Ans. A

Explanation:

- The “Maratha Military Landscapes of India” was inscribed as India’s 44th UNESCO World Heritage Site on 11 July 2025, at the 47th Session of the World Heritage Committee in Paris. It comprises 12 forts — 11 in Maharashtra (Salher, Shivneri, Lohgad, Khanderi, Raigad, Rajgad, Pratapgad, Suvarnadurg, Panhala, Vijaydurg, and Sindhudurg) and one in Tamil Nadu (Gingee Fort) — built, adapted, or expanded by the Marathas under Chhatrapati Shivaji Maharaj and his successors between the late 17th and early 19th centuries.
 - ✓ **Rajgad (1) and Sindhudurg (3) are correctly included** — both part of the 12-fort Maratha listing.
 - ✓ **Gingee Fort (2) is correctly included** — the sole non-Maharashtra fort in the list, notable for extending the network into the Eastern Ghats of Tamil Nadu.
 - ✓ **Kumbhalgarh (4) is not part of this list.** It belongs to an entirely separate UNESCO inscription — the “Hill Forts of Rajasthan” (Chittorgarh, Kumbhalgarh, Ranthambore/Sawai Madhopur, Gagron/Jhalawar, Amber/Jaipur, and Jaisalmer), inscribed back in 2013, as India’s first serial nomination.

Q 114. Ans. A

Explanation:

- **Statement 1 is correct:** Anekantavada (“many-sidedness” or “non-one-sidedness”) is a core Jain doctrine holding that reality is complex and multi-faceted, and can only be fully understood by considering it from multiple, often seemingly contradictory viewpoints. This is closely linked to two other key Jain epistemological concepts — Syadvada (the doctrine of conditional predication, “maybe/perhaps”) and Nayavada (the theory of partial standpoints) — together forming the philosophical basis for Jain religious tolerance and non-absolutism.
- **Statement 2 is correct:** Jainism, like Buddhism and the Charvaka school, is classified as a “heterodox” (nastika) philosophical system precisely because it does not accept the authority of the Vedas. It rejects Vedic ritualism, animal sacrifice, and the concept of a creator God, instead emphasising the doctrine of karma, non-violence (ahimsa), and liberation (moksha) through individual ascetic effort.

- **Statement 3 is incorrect:** Madhyamaka (founded by Nagarjuna, associated with the “Middle Way”/emptiness doctrine) and Yogacara (associated with Asanga and Vasubandhu, emphasising “mind-only” consciousness) are both major philosophical schools of Mahayana Buddhism, not Jainism. Jain philosophy instead developed its own distinct schools of thought centred on Anekantavada, Syadvada, and the doctrine of the Tattvas (fundamental categories of reality), quite separate from these Buddhist schools.

Q 115. Ans. B

Explanation:

- **Rock Edict XIII (the Thirteenth Major Rock Edict) is Ashoka’s most detailed and historically significant inscription regarding the Kalinga War and the profound remorse it triggered in him. In this edict, Ashoka:**
 - ✓ Explicitly describes the conquest of Kalinga in the eighth year of his reign (c. 261 BCE), stating that 1,50,000 people were deported, 1,00,000 were killed in the conflict, and many times that number died from its after-effects.
 - ✓ Expresses deep sorrow (“remorse”) over the immense suffering, death, and displacement caused by the war — describing how the bloodshed weighed heavily on his conscience.
 - ✓ Announces his consequent turn toward Dhamma (Dhamma-vijaya, or “conquest through righteousness”) as the truest form of conquest, renouncing further armed conquest (bherighosha replaced by dhammaghosha — the sound of war-drums replaced by the sound of the Dhamma).
 - ✓ Notably, this edict is absent from copies found in the Kalinga region itself (at Dhauli and Jaugada), where local versions omit the passages describing the massacre and remorse — likely a deliberate administrative choice to avoid distressing/alarming the very population it had subjugated.
- **Additional Facts:**
 - ✓ **Rock Edict X (Option 1):** Deals with Ashoka’s rejection of fame and glory in favour of Dhamma, not specifically the Kalinga War narrative.
 - ✓ **Sarnath Pillar Edict (Option 3):** Primarily addresses the schism (splits) within the Buddhist Sangha, warning against those who create divisions in the monastic order — unrelated to Kalinga.

- ✓ **Arthashastra (Option 4):** Kautilya's treatise on statecraft and political economy, composed during the earlier Mauryan period (under Chandragupta Maurya) — an entirely separate text, not an Ashokan inscription at all.

Q 116. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Chanhudaro, located in Sindh, is a well-known inland industrial centre, notable for its bead-making, seal-carving, and metalworking workshops — and unusually, it lacks a citadel, unlike most major Harappan cities.
- **Pair 2 is correctly matched:** Sutkagendor, situated near the Dasht River close to the Pakistan-Iran border on the Makran coast, is identified as a port settlement, marking the westernmost extent of the Indus Civilization and likely facilitating maritime trade with Mesopotamia/the Persian Gulf.
- **Pair 3 is correctly matched:** Kuntasi, in coastal Gujarat (Rajkot district), functioned as a port-cum-industrial site, with evidence of craft production alongside its coastal trading role.
- **Pair 4 is incorrectly matched:** Balakot is not an "inland agricultural centre" — it is a coastal site near the Arabian Sea (Sonmiani Bay, Pakistan), specifically renowned as a specialised shell-working centre (alongside Nageshwar in Gujarat), where archaeologists found extensive evidence of bangle-making and other shell-craft workshops using bivalve and conch shells.

Q 117. Ans. B

Explanation:

- **Pair 1 is incorrectly matched:** Vikramankadevacharita, an 18-canto Sanskrit epic (mahakavya) composed around 1085 CE, was written by the Kashmiri poet Bilhana, glorifying the life and achievements of the Western Chalukya king Vikramaditya VI (r. 1076–1126 CE), whose court he served after being honoured with the title "Vidyapati." Harshavardhana, by contrast, was a 7th-century emperor and himself an author — credited with the plays Ratnavali, Priyadarshika, and Nagananda — not the composer of this work about a much later Chalukya king.
- **Pair 2 is correctly matched:** Kumarapalacharita (also known as Dvyashraya Kavya) was composed by Hemachandra, the renowned Jain scholar-monk, chronicling the life of the Chalukya (Solanki) king Kumarapala of Gujarat, his patron.
- **Pair 3 is correctly matched:** Rajatarangini, a

12th-century historical chronicle of the kings of Kashmir, was composed by Kalhana — widely regarded as one of the earliest works of systematic historical writing in India.

Q 118. Ans. A

Explanation:

- **Avanti (westernmost):** Centred around the Malwa region of present-day Madhya Pradesh, with its capital at Ujjain (later also Mahishmati) — the westernmost of the four, straddling a key north-south trade route.
- **Vatsa:** Located around present-day Prayagraj (Allahabad), Uttar Pradesh, along the Yamuna River, with its capital at Kaushambi — east of Avanti but west of Kosala.
- **Kosala:** Situated in present-day eastern Uttar Pradesh, along the Sarayu/Ghaghara river system, with its capital variously identified as Shravasti or Ayodhya — further east than Vatsa.
- **Magadha (easternmost):** Located in present-day southern Bihar, with its capitals at Rajagriha (Rajgir) and later Pataliputra — the easternmost of the four, eventually becoming the nucleus of the Mauryan and Gupta empires.

Q 119. Ans. B

Explanation:

- Following the sack of Delhi by Timur in 1398, Khizr Khan was appointed as Timur's deputy in Punjab (Multan), with authority over Lahore, Dipalpur, Multan, and Upper Sindh. Earlier, Khizr Khan had served as the governor of Multan under Firuz Shah Tughlaq, but was expelled in 1395. Timur's invasion and patronage enabled him to re-establish his political influence.
- With Timurid support, Khizr Khan gradually consolidated his power, defeated rivals such as Daulat Khan Lodi and Mallu Iqbal Khan, and captured Delhi in 1414, establishing the Sayyid Dynasty, the fourth of the five ruling dynasties of the Delhi Sultanate. Out of respect for (or fear of) Timur, Khizr Khan never adopted the title of "Sultan." Instead, he used titles such as Rayat-i-Ala ("the exalted subject") and Masnad-i-Aali, and continued issuing coins in the name of the earlier Tughlaq rulers.

Q 120. Ans. B

Explanation:

- Under Sher Shah Suri's efficient administrative system, customs duties across his entire empire

were levied at only two designated points, in order to eliminate the vexatious multiple internal tolls that had previously choked trade at every road, ferry, town, and village:

- ✓ **Sikrighali** (also called “Garhi” in Abbas Khan Sarwani’s *Tarikh-i-Sher Shahi*) — located at the border between Bengal and Bihar — where duties were collected on all goods manufactured in Bengal or imported into it from outside.
- ✓ **The Indus River** — where duties were collected on goods arriving from West and Central Asia (via Khurasan).
- A second duty was then levied only once more, at the actual point of sale, with no other authority permitted to impose customs anywhere else along the route. This system dramatically boosted trade and commerce by removing the burden of repeated taxation at every checkpoint, and local village headmen (muqaddams) and zamindars were held personally responsible for compensating merchants for any losses due to theft within their jurisdiction.

Q 121. Ans. C

Explanation:

- **Statement 1 is correct:** The Amara-Nayakas were military commanders granted territories (amaram) by the Vijayanagara king (raya) in return for maintaining a stipulated contingent of troops, horses, and elephants for the king’s service. They collected taxes and dues from peasants, craftspersons, and traders in their assigned territory, retaining a portion for personal use and military upkeep, while sending tribute (and often making personal appearances, especially during festivals like Mahanavami) to the imperial treasury.
- **Statement 2 is correct:** Historians (notably Burton Stein) have long drawn a close comparison between the Amara-Nayaka system and the Iqta system of the Delhi Sultanate, since both involved assigning revenue-bearing territories to military commanders in exchange for troops and service, with the central ruler retaining the power to transfer assignees between territories. However, a key point of divergence was that, over time — particularly from the mid-16th century onward and accelerating dramatically after the Battle of Talikota (1565), which shattered central Vijayanagara authority — many Amara-Nayakas entrenched themselves in their territories, passed control down within their families, and eventually established fully hereditary,

quasi-independent Nayaka kingdoms (such as those at Madurai, Tanjore, and Gingee). This stands in contrast to the Iqta system, which was designed to remain formally non-hereditary and transferable at the Sultan’s discretion (even though, in practice, Iqtas too sometimes drifted toward hereditary tendencies under weaker Sultans).

Q 122. Ans. B

Explanation:

- **1 is correct:** Lord William Bentinck launched a systematic campaign to suppress Thuggee (organized gangs of ritualistic robber-murderers) starting in 1830, led by Captain William Sleeman, who was appointed to head the Thuggee and Dacoity Department. This campaign, through mass arrests, trials, and the “approver” system (informants), effectively broke the organized Thuggee networks by the mid-1830s.
- **2 is correct:** The Charter Act of 1833 was passed during Bentinck’s tenure, and under this Act, he became India’s first Governor-General of India (upgraded from the earlier title of “Governor-General of Bengal”). The Act also ended the East India Company’s remaining commercial functions, converting it into a purely administrative body, and opened Company service to Indians in principle (Section 87), and centralised legislative power in the Governor-General-in-Council.
- **3 is correct:** The English Education Act of 1835, based on Macaulay’s Minute on Education (1835), was passed during Bentinck’s tenure. It made English the medium of instruction for higher education and the official language of government business, redirecting the Company’s education funds toward English-language, Western-style education over traditional Oriental (Sanskrit/Persian/Arabic) learning.
- **4 is incorrect:** The Permanent Settlement of Bengal was introduced by Lord Cornwallis in 1793 — over three decades before Bentinck’s tenure even began (1828–1835). This is a clear chronological mismatch.

Q 123. Ans. B

Explanation:

- (a). **Taayyuni (Taiyuni) Movement → 3 (Karamat Ali Jaunpuri):** Founded by Karamat Ali Jaunpuri (born Muhammad Ali Jaunpuri, 1800–1873), the Taiyuni movement spread purified, orthodox Islamic teachings across

Bengal and Assam from the 1830s. Unlike the Faraizi movement (led by Haji Shariatullah), which regarded British India as “Dar al-Harb” (a land of war) and even suspended Friday and Eid prayers in protest, the Taiyunis favoured cooperation with British rule.

- (b). **Rahanumai Mazdayasan Sabha → 2 (Dadabhai Naoroji):** Founded in Bombay in 1851 by Naoroji Furdoonji, Dadabhai Naoroji, and S.S. Bengalee, this was a Parsi reform association aimed at restoring the purity of Zoroastrianism and improving the social condition of the Parsi community; its journal was Rast Goftar (“Truth Teller”).
- (c). **Namdhari (Kuka) Movement → 1 (Baba Ram Singh):** Founded by Baba Ram Singh in Punjab in 1857, this movement combined religious reform (opposing caste distinctions, idol worship, and expensive rituals) with early anti-British political overtones, including calls to boycott British goods and courts.
- (d). **Radhaswami Movement → 4 (Shiv Dayal Saheb, “Swamiji Maharaj”):** Founded in Agra in 1861 by Shiv Dayal Saheb, emphasising a spiritual guru-based path to liberation, blending elements of Sant and Sufi traditions.

Q 124. Ans. C

Explanation:

- **Option 1 was an objective:** The Satyagraha protested India’s forced involvement in the Second World War without the consent of Indian representatives — the Viceroy, Lord Linlithgow, had unilaterally declared India a belligerent power in September 1939 without consulting Indian political leadership.
- **Option 2 was an objective:** By carefully restricting participation to specifically selected individuals (starting with Vinoba Bhave, followed by Jawaharlal Nehru), Gandhi sought to demonstrate that the Congress’s restraint was a matter of choice and discipline, not weakness or inability to resist — proving the movement’s continued strength and organisational control even while holding back from a full confrontation.
- **Option 3 was NOT an objective:** this is the correct answer: The entire point of “Individual” Satyagraha was to deliberately avoid a mass movement. Gandhi carefully selected a small number of prominent individuals to court arrest one at a time, in a controlled, symbolic manner — precisely to prevent it from escalating into a nationwide mass civil disobedience campaign, which he felt was

premature and could embarrass Britain during a moment of genuine wartime crisis (especially after the Fall of France in mid-1940).

- **Option 4 was an objective:** The calibrated, non-disruptive nature of the protest was meant to keep the door open for negotiation, giving the British Government a further chance to respond to Congress’s demands peacefully — rather than provoking an all-out confrontation, in the hope that Britain might make political concessions.

Q 125. Ans. A

Explanation:

- **Statement 1 is correct:** The Quit India Resolution explicitly demanded an immediate end to British rule in India.
- **Statement 2 is correct:** The resolution called for the formation of a Provisional Government of India after British withdrawal, alongside a declaration that free India would defend itself against all forms of fascism and imperialism.
- **Statement 3 is correct:** The AICC sanctioned a mass civil disobedience movement against British rule, placing Gandhi in overall command — this is precisely what launched the Quit India Movement, accompanied by his famous “Do or Die” call.
- **Statement 4 is incorrect:** The AICC did not accept the Cripps Proposals — quite the opposite. The Cripps Mission had already failed by April 1942, as Congress rejected Sir Stafford Cripps’s draft declaration (which offered only Dominion status after the war, allowed provinces to opt out of the future Indian union, and made no provision for genuine self-government during the war). This failure was, in fact, a key catalyst that led Gandhi to draft the Quit India resolution in the first place — the two events are directly opposed, not aligned.

Q 126. Ans. A

Explanation:

- **Statement 1 is correct:** Village uplift and social reform programs actively campaigned against deep-rooted social evils such as caste oppression to forge greater societal unity.
- **Statement 2 is correct:** Opposition to early marriage was a key part of the progressive social agenda promoted by nationalist workers and volunteer bodies (samitis) to empower women and reform traditional household structures.
- **Statement 3 is incorrect:** The movement did not promote alcohol consumption. On the contrary,

it championed temperance and campaigned heavily against the consumption of alcohol, often organizing intensive picketing of liquor shops as part of its moral and social cleansing programs.

- **Statement 4 is correct:** Campaigns against the dowry system were similarly integrated into village reconstruction efforts to eliminate financial burdens and regressive practices plaguing families.
- **Additional Facts:**
 - ✓ **Atma Shakti (Self-Reliance):** During the Swadeshi Movement (sparked by the Partition of Bengal in 1905), the philosophy of Atma Shakti (self-help and constructive action), notably emphasized by leaders like Rabindranath Tagore, went beyond mere economic boycotts. It aimed at the social, cultural, and economic regeneration of villages through indigenous efforts.

Q 127. Ans. B

Explanation:

- **Photography (Correct):** The popularity and production of Company Paintings, including the regional Patna Kalam (Patna School of Painting), began to decline sharply in the mid-to-late 19th century primarily due to the advent and commercialization of photography. As cameras became more accessible in India, they offered a much faster, cheaper, and mechanically precise medium for capturing portraits, local trades, and daily life scenes—subjects that were previously commissioned from hand-painting ateliers.
- **Other Factors:** While the introduction of lithographic printing (such as Charles D'Oyly's press in Patna) and shifting colonial tastes also contributed to the commercial displacement of hand-drawn works, photography directly superseded the documentary function that these paintings served for European and local elites.

Q 128. Ans. B

Explanation:

- The statement, "Swadeshim during the days of its potency coloured the entire texture of our social and domestic life," was made by the prominent Indian nationalist leader **Surendranath Banerjee**.
- The quote encapsulates the profound and all-encompassing impact of the **Swadeshi Movement**, which was formally launched in 1905 to protest Lord Curzon's partition of Bengal.
- **Additional Facts:**

- ✓ **The "Indian Burke":** Due to his exceptional oratorical skills and deep grounding in liberal political philosophy, Surendranath Banerjee was often referred to as the "Indian Burke."
- ✓ **Indian Association:** In 1876, he founded the Indian Association, which served as a crucial precursor to the Indian National Congress, uniting educated Indians on a common political platform.
- ✓ **Journalistic Legacy:** He served as the editor of *The Bengalee*, an influential newspaper that he used to shape public opinion against regressive colonial policies and articulate moderate nationalist demands.

Q 129. Ans. A

Explanation:

- **Pair 1 is correctly matched:** The Treaty of Allahabad (1765) was signed following the Battle of Buxar by Robert Clive, Mughal Emperor Shah Alam II, and Shuja-ud-Daula (Nawab of Awadh). Under this treaty, the Mughal Emperor formally granted the Diwani rights (rights to collect revenue) of Bengal, Bihar, and Orissa to the East India Company.
- **Pair 2 is correctly matched:** The Treaty of Sugauli (1816) was signed between the East India Company and the King of Nepal, officially ending the Anglo-Nepal War (Gorkha War) of 1814–1816 and defining the boundary line of Nepal.
- **Pair 3 is NOT correctly matched:** The Treaty of Bassein (1802) was a pact signed between the Peshwa Baji Rao II and the British East India Company after the Battle of Pune. It marked a crucial milestone leading into the Second Anglo-Maratha War, not the First.
- **Pair 4 is correctly matched:** The Treaty of Salbai (1782) was signed between the Maratha Empire and the British East India Company, successfully ending the First Anglo-Maratha War and restoring status quo ante (peace and territorial arrangements) for 20 years.

Q 130. Ans. D

Explanation:

- **Pedro Álvarez Cabral (Correct):** In 1500 CE, Pedro Álvarez Cabral led a massive expedition of 13 ships to India following Vasco da Gama's pioneering voyage. He successfully reached Calicut, negotiated trade terms with the Zamorin (the local Hindu ruler), and established the first Portuguese factory (trading post) on Indian soil. However, relations deteriorated quickly due to commercial rivalries with local Arab traders, resulting in an attack on the factory and the death of several Portuguese men.

Q 131. Ans. A

Explanation:

- **Formation of the Interim Government by the Congress (September 2, 1946):** The Interim Government of India was formed on September 2, 1946, from the newly elected Constituent Assembly. Jawaharlal Nehru headed it as the Vice-President of the Executive Council, marking the first major transfer of executive power to Indian hands before formal independence.
- **Announcement of the end of British rule in India by Clement Attlee (February 20, 1947):** British Prime Minister Clement Attlee made a historic statement in the House of Commons on February 20, 1947, declaring that the British would evacuate British India by no later than June 30, 1948, and appointing Lord Mountbatten as the new Viceroy to execute the transfer of power.
- **Announcement of the June Third Plan (June 3, 1947):** Following extensive consultations, Lord Mountbatten formulated the partition and transfer-of-power blueprint known as the Mountbatten Plan or June Third Plan (June 3, 1947), which was formally accepted by both the Indian National Congress and the Muslim League.
- **Introduction of the Indian Independence Bill in the House of Commons (July 4, 1947):** Based on the June Third Plan, the Indian Independence Bill was introduced in the British House of Commons on July 4, 1947. It was swiftly passed (receiving Royal Assent on July 18, 1947), paving the way for the creation of the independent dominions of India and Pakistan on August 15, 1947.

Q 132. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Gopal Krishna Gokhale proposed the Lightning Conductor Theory. According to this view, early Indian nationalist leaders used A.O. Hume as a catalyst (a lightning conductor) to help channel and unite nationalist forces while keeping political agitation within manageable parameters.
- **Pair 2 is incorrectly matched:** The Safety Valve Theory was popularized by the extremist leader Lala Lajpat Rai in his 1916 book *Young India*, where he argued that the INC was started primarily to save the British Empire from danger rather than to win political liberty. It is incorrectly paired with R.P. Dutt here.
- **Pair 3 is incorrectly matched:** The Conspiracy Theory was developed by the Marxist historian R.

Palme Dutt, who built upon the safety valve concept to argue that the Congress was born through a conspiracy to forestall a popular uprising with the complicity of Indian bourgeois leaders. It is incorrectly attributed to Lala Lajpat Rai in this pair.

- **Pair 4 is correctly matched:** R. Palme Dutt is fundamentally associated with the Marxist Theory (and the related conspiracy interpretation) of the INC's foundation, articulated in his classic work *India Today*.

Q 133. Ans. A

Explanation:

- **Pair 1 is correctly matched:** The second session of the Indian National Congress was held in Calcutta in 1886, and it was presided over by Dadabhai Naoroji (the "Grand Old Man of India").
- **Pair 2 is NOT correctly matched:** The Calcutta session of 1890 was presided over by Sir Pherozeshah Mehta, not Dinshaw E. Wacha. Dinshaw E. Wacha served as the INC president later during the Calcutta session of 1901.
- **Pair 3 is correctly matched:** The 1896 Calcutta session was presided over by Rahimtulla M. Sayani.
- **Pair 4 is correctly matched:** The 1911 Calcutta session was presided over by Bishan Narayan Dhar (also spelled Dar).
- **Additional Facts:**
 - ✓ **Historical Significance of 1896:** The Calcutta session under Rahimtulla Sayani is landmarked in history because India's national song, *Vande Mataram* (composed by Bankim Chandra Chattopadhyay), was sung for the first time during a Congress session.
 - ✓ **Historical Significance of 1911:** The Calcutta session under Bishan Narayan Dhar is famous for the first-ever public rendering of *Jana Gana Mana* (composed by Rabindranath Tagore), which later became India's National Anthem.

Q 134. Ans. A

Explanation:

- **Pair 1 is correctly matched:** The Eka Movement (Unity Movement) took place in the northern districts of Awadh (such as Hardoi, Bahraich, and Sitapur) in late 1921 and early 1922. It was organized by backward-caste leaders and notably led by Madari Pasi. The core grievances centered around high rents, illegal evictions, and the oppressive practice of begar (forced or unpaid labor), leading peasants to pledge a refusal to perform forced labor.

- **Pair 2 is NOT correctly matched:** The Mappila Revolt of 1921 in the Malabar region of Kerala was indeed led by figures like Ali Musaliar against oppressive jenmies (landlords) and colonial policies. However, the region operated under Malabar tenancy/Ryotwari-related structures rather than the Permanent Settlement, which was primarily implemented in Bengal, Bihar, and parts of northern Madras. Therefore, the stated demand regarding the Permanent Settlement is historically inaccurate for Malabar.
- **Pair 3 is correctly matched:** The Bardoli Satyagraha of 1928 was a major no-tax campaign launched in Gujarat against a steep 30% land revenue hike by the British administration. It was successfully organized and led by Vallabhbhai Patel, and it was the women of Bardoli who conferred upon him the honorific title “Sardar” (leader).

Q 135. Ans. B

Explanation:

- **Right to peaceful picketing (Accepted):** Lord Irwin agreed to the right to engage in peaceful and non-aggressive picketing of liquor, toddy, and foreign cloth shops as a concession under the pact.
- **Public inquiry into police excesses (Not Accepted):** Lord Irwin flatly rejected the Congress’s demand for an official public inquiry into police brutality and excesses committed during the suppression of the Civil Disobedience Movement.
- **Commutation of Bhagat Singh’s death sentence to life imprisonment (Not Accepted):** Despite repeated appeals by Mahatma Gandhi, the Viceroy refused to commute the death sentences of Bhagat Singh, Sukhdev, and Rajguru, resulting in their execution shortly thereafter on March 23, 1931.
- **Release of political prisoners convicted of violent offences (Not Accepted):** The British government agreed to release only those political prisoners who had been arrested for participating in the Civil Disobedience Movement without being charged with violence. Prisoners convicted of violent crimes or offences were explicitly excluded and kept behind bars.

Q 136. Ans. C

Explanation:

- **India and Pakistan (Correct):** As detailed by Ramachandra Guha in *India After Gandhi*, the sterling balances were the massive financial reserves accumulated by undivided India in London as payment for goods and services supplied to Great Britain during

the Second World War. Following the partition of British India in August 1947, these sterling assets were jointly owned by India and Pakistan, requiring structured negotiations to divide and distribute the funds between the two successor dominions.

● Additional Facts:

- ✓ **The Financial Dispute:** The distribution of the sterling balances became a major flashpoint between New Delhi and Karachi. India temporarily withheld Pakistan’s share (amounting to 55 crore rupees) after the outbreak of the First Kashmir War in late 1947, with leaders like Vallabhbhai Patel arguing against funding an adversary engaged in active military conflict against India.
- ✓ **Mahatma Gandhi’s Final Fast:** The tensions surrounding the delayed transfer of these cash balances to Pakistan formed a critical backdrop to Mahatma Gandhi’s historic fast unto death in January 1948, which he undertook primarily to restore communal harmony and ensure that financial debts and moral obligations to the neighboring dominion were honored.

Q 137. Ans. D

Explanation:

- **Jawaharlal Nehru (Correct):** The immortal lines, “The light has gone out of our lives and there is darkness everywhere...” were spoken by Jawaharlal Nehru in a radio broadcast to the nation on January 30, 1948, immediately following the assassination of Mahatma Gandhi. His poignant address began with these words to capture the profound shock, grief, and sense of loss felt across the country.
- **Additional Facts:**
 - ✓ **The Full Address:** In his broadcast, Nehru went on to say, “The light has gone out of our lives and there is darkness everywhere, and I do not quite know what to tell you or how to say it. Our beloved leader, Bapu as we called him, the father of the nation, is no more.”

Q 138. Ans. A

Explanation:

- **Pair 1 is correctly matched:** The Policy of Ring Fence is associated with Warren Hastings. This defensive policy aimed at creating buffer zones (such as Awadh) to defend the East India Company’s frontiers against direct Marathi and Afghan incursions, ensuring that British territories were protected by shielding them with allied states.

- **Pair 2 is correctly matched:** The Subsidiary Alliance was systematically introduced and aggressively expanded by Lord Wellesley (Governor-General from 1798 to 1805). Under this system, Indian rulers were forced to station British troops within their territories, pay for their maintenance, and surrender control of their foreign policy.
- **Pair 3 is NOT correctly matched:** The Doctrine of Lapse is famously associated with Lord Dalhousie (who served as Governor-General from 1848 to 1856), not Lord Cornwallis. Under Dalhousie's policy, dependent states without a natural male heir were annexed by the Company (e.g., Satara, Jhansi, Nagpur). Lord Cornwallis is instead renowned for the Permanent Settlement of Bengal (1793) and civil/judicial administrative reforms.
- **Pair 2 is correctly matched:** The Charter Act of 1833 marked the final step toward complete financial, legislative, and administrative centralization in British India by designating the Governor-General of Bengal as the Governor-General of India and stripping the Madras and Bombay presidencies of their independent legislative powers.
- **Pair 3 is NOT correctly matched:** The Charter Act of 1853 reduced the number of members of the Court of Directors from 24 to 18 (not 15), out of whom 6 were to be nominated by the British Crown.
- **Pair 4 is correctly matched:** The Charter Act of 1813 explicitly asserted the sovereignty of the Crown over the Indian territories and substantially enlarged and strengthened the supervisory powers of the Board of Control over the East India Company's administration.

Q 139. Ans. A

Explanation:

- **Statement 1 is correct:** Under the Government of India Act, 1935, the Council of States (the proposed upper house at the federal level) was constituted as a permanent body not subject to dissolution, with one-third of its members retiring every three years, mirroring the structural design of modern upper houses.
- **Statement 2 is incorrect:** Women were not granted the right to vote on the same basis as men under the Act. While the franchise was extended to include more women, it remained heavily restricted based on property, educational qualifications, or status as a wife/widow of a qualified voter, resulting in a heavily skewed voter ratio.
- **Statement 3 is incorrect:** The Act introduced bicameral legislatures in only 6 out of the 11 provinces (Bengal, Bombay, Madras, Bihar, United Provinces, and Assam), rather than all provinces.
- **Statement 4 is incorrect:** In a highly unusual setup that departed from standard democratic models, the 1935 Act mandated that the Federal Assembly (lower house) be indirectly elected by the members of the provincial assemblies. Conversely, the members of the Council of States (upper house) representing British India were to be directly elected by a restricted electorate.

Q 140. Ans. A

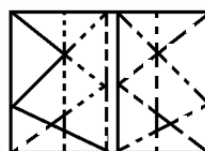
Explanation:

- **Pair 1 is correctly matched:** The Charter Act of 1793 mandated that royal approval (sanction of the British Monarch) was required for the appointment of key high-ranking officials, including the Governor-General, the governors of the presidencies, and the Commander-in-Chief.

Q 141. Ans. D

Explanation:

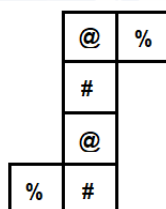
The embedded figure is shown below:



Hence, option D.

Q 142. Ans. C

Explanation:



As, the same type of symbols will appear on the opposite side of paper.

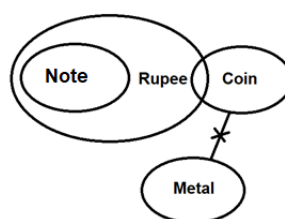
So, '4' will appear opposite to '2'.

Hence, option C.

Q 143. Ans. A

Explanation:

The following figure can be formed:



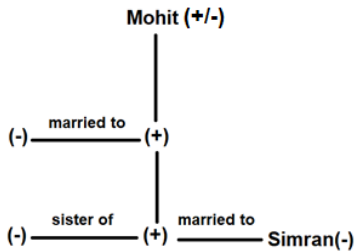
Some Rupee is not Metal as No Coin is Metal.

From the above figure, only conclusion I follow.
Hence, option A.

Q 144. Ans. B

Explanation:

From the information given in the question, we can draw the following family tree:

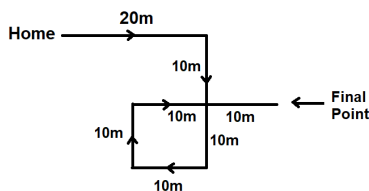


Therefore, Mohit's granddaughter is sister-in-law of Simran.
Hence, option B.

Q 145. Ans. A

Explanation:

Figure:



Clearly,
Geet is in the south-east direction with respect to her home.
Hence, option A.

Q 146. Ans. A

Explanation:

According to the question,
The first n multiples of 8 are:

8, 16, 24, ..., $8n$

Arithmetic mean = (First term + Last term) / 2

$$\Rightarrow (8 + 8n) / 2 = 52$$

$$\Rightarrow 8 + 8n = 104$$

$$\Rightarrow 8n = 96$$

$$\Rightarrow n = 12$$

Therefore, $n = 12$

Q 147. Ans. C

Explanation:

According to the question,

$$0.75 \text{ is a rational number because } 0.75 = \frac{75}{100} = \frac{3}{4}$$

$$\frac{9}{11} \text{ is a rational number.}$$

25 is an integer; therefore, it is also a rational number.

However, $\sqrt{13}$ is an irrational number because 13 is not a perfect square and cannot be expressed in the form p/q .

Q 148. Ans. C

Explanation:

● Given:

- ✓ Principal (P) = ₹40,000
- ✓ Interest rate = 10%
- ✓ Interest for the first year = $40,000 \times 10\% = 4,000$
- ✓ Amount at the end of the first year = $40,000 + 4,000 = 44,000$

● Second year:

- ✓ Interest rate = $10\% + 4\% = 14\%$
- ✓ Interest for the second year = $44,000 \times 14\% = 6,160$
- ✓ Total compound interest (over 2 years) = $4,000 + 6,160 = 10,160$

Q 149. Ans. C

Explanation:

According to the question,
Let the number of boys and girls be $5x$ and $6x$, respectively.

According to the question,

$$\frac{5x - 8}{6x - 4} = \frac{21}{26}$$

$$\Rightarrow 26(5x - 8) = 21(6x - 4)$$

$$\Rightarrow 130x - 208 = 126x - 84$$

$$\Rightarrow 4x = 124$$

$$\Rightarrow x = 31$$

Therefore,

Number of boys = $5 \times 31 = 155$

Number of girls = $6 \times 31 = 186$

Total number of students = $155 + 186 = 341$

Thus, there were 341 students in the class initially.

Q 150. Ans. C

Explanation:

Ratio of boys and girls = 21 : 16

Let the number of boys = 210

Let the number of girls = 160

Boys selected for the sports competition =

$$210 \times 60\% = 126$$

Girls selected for the sports competition =

$$160 \times 75\% = 120$$

Total students selected = $126 + 120 = 246$

Total students = $210 + 160 = 370$

Students not selected for the competition =

$$370 - 246 = 124$$

$$\text{Required percentage} = \frac{124}{370} \times 100\% = 33.5\%$$

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